

Fast and Stable Virtual Human Video Generation: A Survey

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Abstract Virtual-human video generation must deliver timely output while preserving identity, appearance, and motion over sustained use. These requirements interact through representation, temporal state, and serving design, yet published results use different measurement boundaries. This survey examines computational efficiency and temporal reliability through two lenses, *Fast* and *Stable*. The evidence centers on talking portraits, with selected upper- and full-body systems. We organize methods into GAN, diffusion, visual-autoregressive, and rendering routes, using explicit hybrid-assignment rules and independent architectural annotations. A shared failure vocabulary distinguishes appearance drift, discontinuity, control error, geometric artifacts, and service failure; five state-management patterns describe mechanisms rather than guarantees. Three within-source analyses show that decoder acceleration can trade identity and temporal quality for throughput, serving parallelism can improve output rate without shortening startup, and short-horizon ablations establish narrower claims than long-session reliability. Reusable appearance reduces repeated synthesis but leaves controller and rendering failures to be evaluated. A versioned evidence supplement preserves source editions and audit status, while a worked evaluation protocol separates absolute quality, temporal change, intended control, and system cost. This organization supports deployment decisions without treating heterogeneous source measurements as a universal ranking.

1 Introduction

Virtual-human video generation aims to synthesize a human subject whose appearance and behavior follow

specified driving signals, such as speech, text, pose, expression, or video. The output may show a face, an upper body, or a full body, while the subject may be specified by one reference image, an enrollment recording, or a learned identity representation. Across these settings, the central requirements are to preserve the intended appearance and reproduce the motion conveyed by the input [123, 167, 194, 215].

Practical deployment introduces two further requirements. The first is computational efficiency: generation must fit the available compute and latency budget, particularly when serving many users or following a live signal [58, 76, 192]. The second is long-horizon stability: identity, appearance, synchronization, and motion should remain consistent as generation continues [102, 152]. We refer to these perspectives as *Fast* and *Stable*. Short-clip visual quality alone establishes neither property. We therefore examine how each family of methods reduces computation, maintains temporal consistency, and trades these objectives against representation quality and deployment cost.

Figure 1 illustrates the distinction: delayed output limits responsiveness, whereas unintended changes in identity or scene limit continuity. A useful evaluation must measure both, while allowing changes explicitly requested by the driving signal.

Figure 2 places representative methods on the four ownership routes over time. Recent work reduces different costs. Distillation can reduce the number of diffusion evaluations [116, 160, 205, 206]. Caching can reuse computation across denoising steps or adjacent frames [107, 118, 121]. Autoregressive models can generate cross-frame visual state sequentially and reuse cached history [26, 93, 191]. Subject-specific renderers can reuse an enrolled scene or reference representation across frames [58, 59, 188, 192]. These mechanisms change different cost terms and should not be treated as equiv-

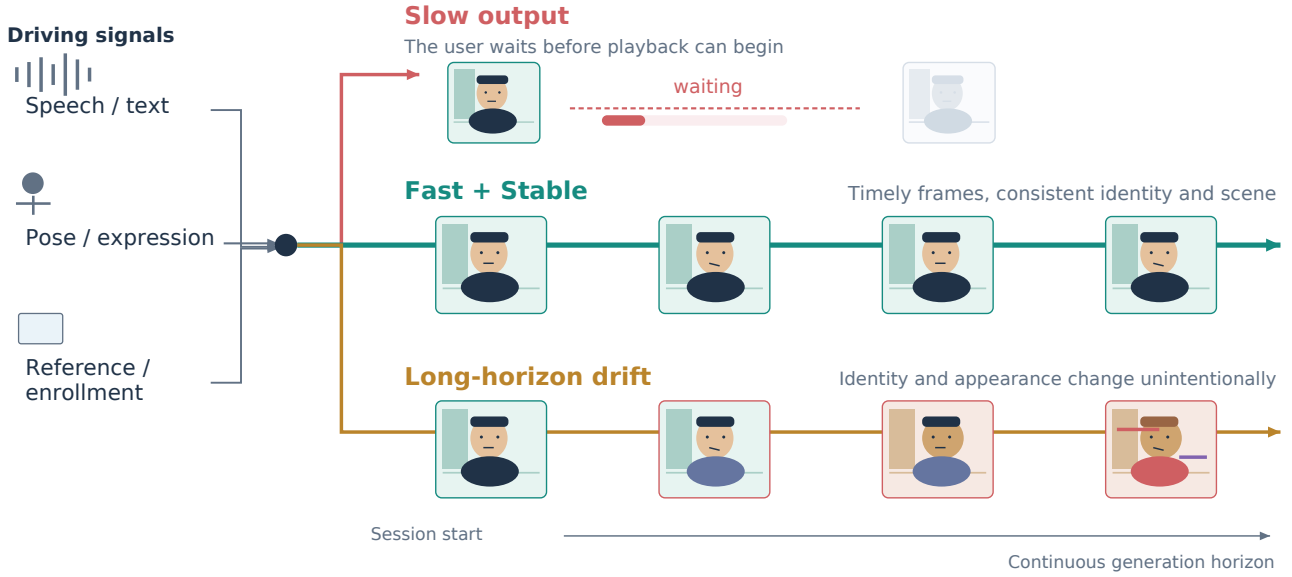


Fig. 1 Fast and Stable are distinct, coupled requirements for virtual-human generation. The central path emits timely frames while preserving identity and scene appearance. The upper branch delays playback; the lower branch illustrates unintended identity and appearance drift during continuous generation. Frame glyphs are schematic.

Route	Earlier mechanism	Intermediate development	Recent streaming mechanism
GAN	Wav2Lip+GAN (2020): regional face synthesis [138]	Live Speech Portraits (2021): compact audio-driven motion [113]	MuseTalk (2024): one-call lower-face latent prediction [218]
Diffusion	EMO (2024): expressive audio-conditioned video [167]	FADA (2025): denoiser-call reduction [229]	Live Avatar (2025): few-step sampling, noisy KV, pipeline parallelism [76]
Visual AR	Body of Her (2024): causal frame state and local sampling [4]	MIDAS (2025): compressed frame representations [26]	EARTalking (2026): sink, sliding cache, cyclic positions [187]
Rendering	AD-NeRF (2021): audio-conditioned fields [59]	StyleAvatar (2023): rendered UV features and adversarial decoding [178]	LivePortrait (2024): reusable source volume and learned warping [58]

Fig. 2 Selected route milestones and their mechanisms. Dates indicate the first public edition, not a priority claim. This selection is illustrative rather than a publication census.

alent: a measured frame rate does not by itself establish first-output latency or long-session stability, and a local speedup can move a bottleneck to the decoder, the condition encoder, or the serving system rather than remove it.

These developments motivate an organization based on the component responsible for generated appearance, which we call the *visual owner*. Systems with similar outputs can have different visual owners and computational costs, whereas systems covering different body regions can share the same acceleration mechanism. Identifying this component provides a common starting point for discussing efficiency and stability within each method family.

1.1 Scope: Fast and Stable

The main subjects of this survey are systems that produce human-centered RGB video and introduce mechanisms for reducing runtime cost or supporting extended generation. Our evidence base centers on talking portraits, with selected upper- and full-body systems included to examine representation and deployment differences. We cover generalizable and subject-specific methods without claiming exhaustive coverage of whole-body generation. Render-based systems are included when they contribute to efficient appearance construction or frame production, for example through neural radiance fields (NeRFs) or 3D Gaussian splatting (3DGS) [83, 126]. Methods whose final output consists only of a mesh, motion coefficients, or vertex trajectories are discussed as

supporting components when they clarify the design of an RGB-generation pipeline.

We distinguish the systems under comparison from foundational and contextual references. Foundational methods explain the origin of a representation or mechanism; contextual references clarify an interface, such as motion prediction followed by rendering. Neither role implies inclusion in a route’s deployment comparison. Inclusion in a primary route requires sufficient architectural evidence to identify the visual owner. Works with undisclosed ownership, or with a visual synthesis mechanism outside the four families considered here, are omitted from the route comparisons. Missing timing or stability measurements do not prevent an otherwise supported architectural assignment; such measurements are marked not reported (NR).

The following two cross-cutting lenses organize the analysis.

Fast concerns the computation and time required to produce visible output. We examine representation size, model-call count, cost per call, computation reuse, and decoding or rendering overhead. Throughput and first-output latency are recorded separately under the source’s hardware and timing conditions.

Stable concerns consistency over an evaluated continuous duration. We examine identity and appearance drift together with the memory, training, and recovery mechanisms intended to control it. A long demonstration establishes an observed duration; quantitative stability additionally requires measurements over that duration.

Interactivity is a related but distinct property: a system can respond to new conditions during generation, such as a dialogue turn, interruption, or change of pose [76, 102, 197, 211, 213]. This capability does not establish stability over a long session; an interactive system may operate briefly, reset frequently, or accumulate drift. We include interactive systems when their generation mechanisms fall within the stated scope and examine their efficiency and stability evidence separately. Interaction behavior has its own surveys and evaluation protocols [7, 14, 46, 209] and is not ranked here.

Fast and Stable are distinct evaluation axes whose design choices can be coupled. A fast method may drift within a minute; a stable long-horizon method may run far below playback speed. We accept a joint claim only when each stated property has independent evidence. Section 2 defines the terms and evidence boundaries used throughout the survey.

1.2 Literature Selection and Evidence Extraction

This is a narrative survey with a retrospective evidence audit dated 9 October 2026. The auditable starting mate-

rial comprises the draft bibliography, the accompanying method catalogue, a locally supplied paper collection, and two supplementary AR collections recorded in the catalogue provenance. We followed those records to primary papers and official project material, prioritizing the architectures and measurements used in the comparison tables. We do not have a prospectively logged database-query history and therefore do not describe the collection as a systematic search or infer field-wide prevalence from its counts. The audit date bounds this snapshot; it is not a claim that every work available by that date was retrieved.

We distinguish primary RGB-generation systems, foundational references, and contextual controllers or interfaces. Architectural inclusion follows Section 2.3. The detailed comparison cohort covers the 57 systems in the submitted route-evidence tables, with corrections to their assignment and measurements; the compact tables emphasize distinct acceleration mechanisms, hybrid boundaries, or available timing and temporal evidence. This retrospective cohort rule makes selection reproducible without implying that the sample is exhaustive or statistically representative. Works cited only to explain a foundation or an interface do not contribute to route-level evidence counts.

Snapshot VH-2026-10-09 separates the 57 inherited system records, whose fields have not all been rechecked, from 22 source-checked configuration or architecture records covering 12 systems. The focused analyses in Section 7 use the checked records. Each checked configuration records a source edition, source location, reported value, and measurement boundary. Archival publications and preprints are retained as distinct editions; a later preprint does not silently replace the edition behind an earlier value. Supplements and official technical documentation are separately identified. Successive system releases are linked rather than counted as independent architectural confirmations. The audited Live Avatar v6 edition, for example, reports H100 hardware; the submitted text used the earlier H800 label. We adopt the pinned edition and preserve the change in the ledger.

Missingness has four distinct meanings: **NR**, a field not reported in the checked source; **U**, a source unavailable for verification; **C**, an unresolved conflict; and **NA**, a field inapplicable to the stated task or boundary. Records carried forward without a fresh source check are additionally marked as such. NR does not mean that no public evidence exists. Conflicting values remain separate until their edition or configuration explains the discrepancy. Ambiguous route assignments are resolved from the inference graph and geometric appearance path; unsupported assignments are omitted from the primary comparison.

The **companion catalogue** exposes the frozen evidence sheet and the maintained reading collection. The 164-record catalogue is a curated reading resource with primary, foundational, and contextual entries. It is not a one-to-one index of the paper’s bibliography; source editions, aliases, and supporting citations account for different roles in the two collections.

1.3 Relation to Existing Surveys

Prior surveys provide complementary views of the area. Work on efficient generative modeling reviews distillation, caching, attention, and inference optimization [151]. Long-video surveys organize temporal extension around generation paradigms and resource or quality challenges [94]. Human-centered surveys organize talking heads, portrait animation, and human video by task or representation [123, 144, 148, 194]. Graphics-oriented reviews cover avatar construction and learned appearance priors [237]. Recent surveys consider interactive generative video, organized around the interaction loop [209], and real-time interactive avatars, organized by method family and by production axes such as rendering latency, anatomical coverage, and personalization [46].

Efficient-diffusion surveys organize algorithmic and system acceleration, while long-video surveys examine temporal extension and its resource demands [94, 151]. Within the virtual-human setting, the closest treatment reviews real-time interactive avatars and comes near our concerns by making rendering latency one of its organizing axes, alongside anatomical coverage, personalization, and input modality [46]. Its grouping is by representation family, however, and it treats long-horizon identity drift as an open challenge rather than a systematized design axis. The remaining reference points organize differently again: a task taxonomy groups systems with different inference loops and temporal state, an acceleration taxonomy groups caches that reuse different objects [123, 151, 194], and an interaction taxonomy foregrounds the dialogue loop rather than the generation cost [209]. We therefore ask three questions. Which component owns generated appearance? How is that route made fast? How is it kept stable over a long horizon? Answering these per route, through an ownership-based classification and route-specific efficiency and drift-control guidance, is what distinguishes this survey.

1.4 Ownership Taxonomy

Figure 3 assigns each included system to one of four routes by identifying its visual owner. The rules are

applied in order: render-based, autoregressive, diffusion-based, then GAN-based. This convention resolves hybrids, such as an autoregressive motion controller driving a subject-specific renderer. It is not intended to classify every possible video generator. Section 2.3 defines the rules and the evidence required for an assignment.

- **GAN-based methods** (Section 3) use an adversarially trained feed-forward generator for dense or regional imagery [54, 81, 138, 163, 218]. Regional prediction and compact control reduce per-frame work, while temporal consistency depends on the generator’s conditioning and any temporal components.
- **Diffusion-based methods** (Section 4) use diffusion or flow matching to generate dense pixels, visual latents, or visual chunks [65, 68, 106, 167]. Acceleration reduces iterative calls, per-call cost, or repeated work; stability mechanisms include reference anchors, bounded memory, and training with generated history.
- **Autoregressive methods** (Section 5) predict evolving visual tokens or latents from previous visual state [4, 26, 187, 191]. Token volume and serial prediction determine much of the cost, while retained history can support continuity and propagate errors.
- **Render-based methods** (Section 6) reuse a subject-specific mesh, neural field, Gaussian set, or identity-conditioned warp [58, 83, 99, 126, 192]. Reused appearance can reduce repeated synthesis and anchor identity, with enrollment, tracking, deformation, and compositing contributing additional costs and failure modes.

Driving modality, anatomical coverage, personalization, representation, and deployment mode are comparison axes rather than additional routes. Secondary diffusion, autoregressive, GAN, or rendering modules are recorded as boundary components and are not double-counted in route totals.

1.5 Contributions and Main Findings

The survey contributes an explicit hybrid-assignment procedure, a shared failure and mechanism vocabulary, and a source-linked evidence supplement. Its central findings are developed in Section 7:

1. **Remaining stages determine the value of a local speedup.** PersonaLive’s decoder ablation exposes a measurable speed–quality trade-off after visual generation is accelerated.
2. **Serving topology changes throughput and startup differently.** Live Avatar’s configurations distinguish model-call reduction, extra GPUs, decoder parallelism, and kernel optimization.

Table 1 Relation to two complementary reference surveys. Entries describe their organizing emphasis, not an assertion that unlisted topics are absent.

Dimension	Human Motion Video Generation [194]	Efficient Diffusion Models [151]	This survey
Scope and index	Human motion video; five stages from planning to output	General diffusion; algorithms, systems, and frameworks	Portrait-centered RGB generation; four primary routes with independent annotations
Hybrid explanation	Task, driving modality, and stage interfaces	Composition of acceleration techniques	Inference-graph and geometric-transport tests with worked cases
Cost emphasis	Generation tasks and their component methods	Training and inference efficiency	Enrollment, component work, startup, and sustained output
Temporal emphasis	Motion quality and temporal generation challenges	Sampling and system efficiency mechanisms	Shared failures; bounded memory, positions, history training, and boundary transfer
Evidence resource	Dataset/metric survey and a common nine-method TikTok comparison	Mechanism hierarchy and published efficiency results	Pinned source configurations, three within-source analyses, and a proposed session protocol

Identify the visual owner; stop at the first satisfied rule.

1. Does a subject-specific scene or identity-conditioned warp carry appearance before rendering? ↓ No: test the next rule.	$\xrightarrow{\text{Yes}}$	Render-based
2. Does native autoregression predict evolving dense or discrete visual state from prior visual state? ↓ No: test the next rule.	$\xrightarrow{\text{Yes}}$	Autoregressive
3. Does diffusion or flow matching generate dense pixels, visual latents, or visual chunks? ↓ No: test the next rule.	$\xrightarrow{\text{Yes}}$	Diffusion-based
4. Does an adversarially trained feed-forward generator produce the final dense or regional imagery?	$\xrightarrow{\text{Yes}}$	GAN-based

Scope and evidence: A method enters the comparison only when its documented architecture supports one of these assignments. Otherwise it is omitted from the route comparisons. Auxiliary controllers and generic VAE decoders do not determine the route.

Fig. 3 Assignment to the four visual-generation routes. For methods meeting the scope and evidence requirements, the first satisfied rule determines the primary route. The order resolves hybrid cases without ranking quality or efficiency. Methods lacking a supported assignment are excluded from route comparisons. AR denotes autoregressive; VAE denotes variational autoencoder.

3. **Memory management and demonstrated stability are separate results.** A fixed cache gives a resource bound; history and anchor ablations test output failures over specified horizons.
4. **Appearance reuse relocates the stability problem.** Source-conditioned rendering anchors identity, while motion, tracking, deformation, and composition remain separate failure paths.

The first two findings use within-source measurements. The latter two combine architectural deductions with bounded empirical examples. Proposed transfers are marked as hypotheses. We provide an implementable evaluation protocol, without claiming to have rerun the models on common hardware.

1.6 Organization

Section 2 defines the task, route ownership, and the cost and measurement language. Sections 3–6 review the four routes under common Fast and Stable lenses, each ending with route-specific design guidance. Section 7 develops the evidence-backed findings. Section 8 examines datasets, metrics, and a worked evaluation protocol. Section 9 develops cross-route design guidance and future directions.

2 Preliminaries and Common Evaluation Language

This section introduces a common formulation for the systems reviewed below. We first distinguish the generated representation from its decoder or renderer, then define route assignment and the measurements used to compare efficiency and stability.

2.1 Problem Formulation and System Boundary

Let $V = (I_1, \dots, I_T)$ denote the generated human video. A system may use enrollment evidence e (e.g., subject video, calibrated views, or a pre-built avatar), runtime conditions $c_{1:T}$ (e.g., audio, text, pose, expression, or a driving video), and persistent deployment state h_t . We write

$$(I_{t:t+k-1}, h_{t+k}) = \mathcal{G}(e, c_{\mathcal{V}_t}, h_t; \theta), \quad (1)$$

where k is the emitted frame or chunk length and $\mathcal{V}_t$ is the portion of the conditioning stream consumed at emission time. This formulation covers one-shot, subject-specific, and generalizable systems without assuming that a reference image is supplied at every runtime step.

Three output requirements recur across tasks. *Identity fidelity* asks whether the intended subject remains recognizable; *driving fidelity* asks whether speech, expression, pose, gaze, and gesture follow their controls; and *temporal fidelity* asks whether motion and appearance remain coherent across emitted units. Their measurement depends on anatomy, modality, personalization, and the availability of a paired reference, so no single metric is valid for every method.

For analysis, we expose an intermediate representation r and a decoding or rendering operation D :

$$\begin{aligned} (r_{t:t+k-1}, h_{t+k}) &= G(e, c_{\mathcal{V}_t}, h_t), \\ I_{t:t+k-1} &= D(r_{t:t+k-1}). \end{aligned} \quad (2)$$

Equation (2) is a bookkeeping view, not a mandatory two-network architecture. In some systems, G and D are fused; in others, e is compiled into a subject-specific scene or the parameters of D before runtime. Any subject evidence consumed at decoding is therefore included in r or in those precompiled parameters under this notation. The distinction remains useful because it identifies the representation and state-update owner, the decoder or renderer boundary, and the cost paid during enrollment versus repeated inference.

Figure 4 makes this accounting explicit: G owns the generated representation and persistent-state update, while D maps that representation to observable frames. The clocks are separate reporting boundaries, so a steady-state throughput result cannot substitute for enrollment, first-output, or session-horizon evidence.

2.2 Efficiency and Stability Evidence

The Fast and Stable perspectives require different evidence. Efficiency is assessed from the computation and measured timing of a declared pipeline, including conditioning, synthesis, decoding, and system overhead. Stability is assessed from changes in identity, appearance,

synchronization, and motion over a declared generation horizon, together with the state and recovery policy used during that run.

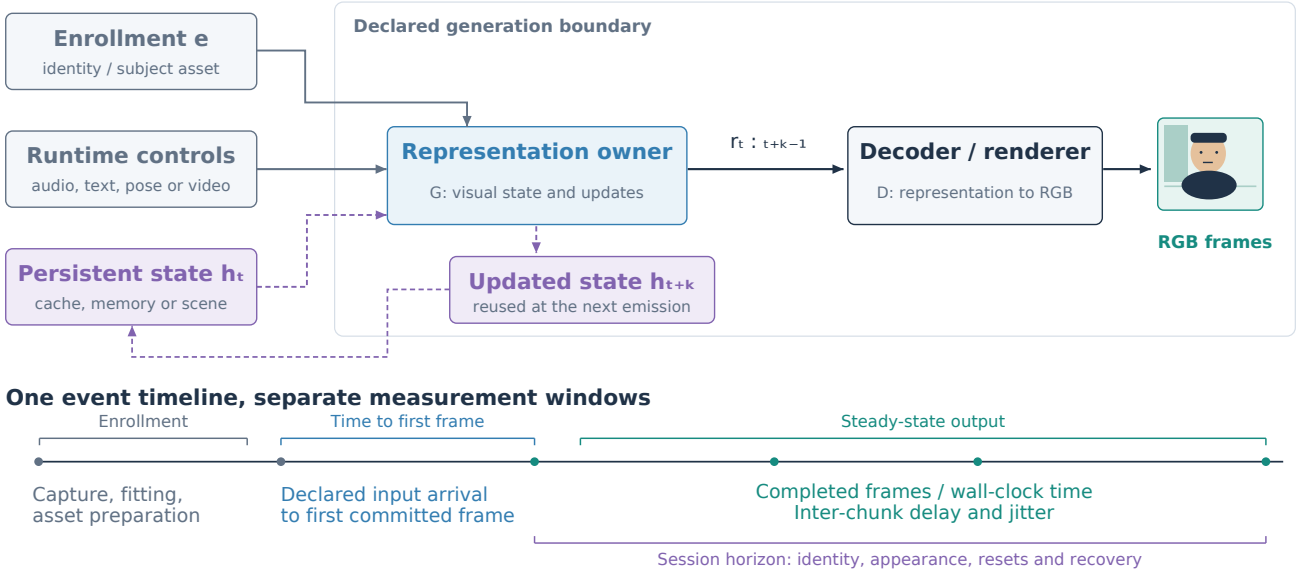
For long-horizon results, we distinguish an author’s duration claim, a qualitative demonstration, and quantitative measurements over time. These support different conclusions. A bounded cache establishes a bound on stored context, while drift must be measured from the generated output. Similarly, sequential emission and responsiveness to new input describe operating capabilities; neither establishes long-horizon stability. Enrollment duration is recorded separately from generation duration.

2.3 Ordered Route Ownership

For a system within the scope of Section 1.1, we inspect the documented representation, its temporal update, and its path to RGB. We then apply the following rules in order and assign the first matching route (Figure 3). This precedence resolves hybrid architectures; it does not rank quality or efficiency.

1. **Render-based.** A subject-specific explicit or neural scene, including a mesh, neural field, Gaussian set, or identity-conditioned warp, owns appearance and is queried or updated before rendering. A learned model that predicts compact motion, geometry, deformation, or Gaussian attributes remains a controller.
2. **Autoregressive.** Otherwise, a native autoregressive model owns cross-frame dense or discrete visual state, such as visual tokens or visual latents. A downstream diffusion process or renderer may be a local emission head, but it does not own the evolving visual sequence.
3. **Diffusion-based.** Otherwise, diffusion or flow matching owns dense pixels, dense visual latents, or sequential dense visual chunks. A generic variational autoencoder (VAE) decoder does not displace this ownership.
4. **GAN-based.** Otherwise, an adversarially trained feed-forward generator owns final dense or regional visual synthesis. Adversarial loss attached only as a regularizer is insufficient.

An assignment requires architectural evidence from the primary paper or its accompanying technical description. If that evidence is insufficient, or none of the four rules applies, the work is excluded from route comparisons rather than assigned an additional category. The taxonomy is therefore an organization of the reviewed method families, not an exhaustive partition of all video synthesis algorithms.



Every clock reports its included components. G and D may be fused; throughput cannot substitute for TTFF or stability.

Fig. 4 Analytical generation boundary and independent reporting clocks. The representation owner G updates visual state; decoder or renderer D produces observable RGB frames. Enrollment, time to first frame, steady-state output, and the evaluated session horizon retain their own measurement boundaries. G and D need not be separate networks.

Supporting modules are identified by their role in the complete pipeline. A diffusion motion controller driving a subject-specific renderer remains part of a render-based system; an autoregressive dialogue model emitting pose controls does not become a visual AR method. Missing performance fields are marked NR independently of this architectural decision. Historical and contextual examples are identified explicitly and do not supply deployment evidence for another route.

An operational test for hybrid systems. The primary route is an editorial index, not a claim that adversarial learning, diffusion, autoregression, and rendering are disjoint properties. We separately record appearance representation, temporal factorization, training/sampling mechanism, and personalization. The visual owner can differ from both the latency bottleneck and the component that causes a temporal failure.

For the AR–diffusion boundary, inspect where the noisy current sample enters the inference graph. A visual AR backbone computes a next-unit state $h_t = F(c_{\leq t}, z_{< t})$ independently of the local sampling noise; a head then samples z_t conditioned on that fixed state. The sampled unit is committed to the next backbone step. A chunkwise diffusion backbone instead repeatedly evaluates $v_{\theta}(z_t^{(s)}, s, z_{< t}, c)$ while denoising the current chunk, with the current noisy sample and noise level entering the temporal visual backbone. Both constructions factor generation causally. We index the former

as visual AR and the latter as diffusion, retaining local iterative sampling in the cost ledger in either case.

For the rendering boundary, inspect whether an explicit identity-specific spatial representation is queried or transported under geometric control before neural image decoding. Learned decoding and adversarial losses do not negate that path. Applying this rule consistently places StyleAvatar, with its rendered facial UV features and compositional decoder, in the render route alongside LivePortrait’s source-volume warping. StyleAvatar remains a useful contextual example of adversarial synthesis in Section 3.4; it is counted only once in route comparisons. Static features without a documented geometric query or transport path would not alone satisfy this test.

2.4 Route-Specific Cost Structures

The same visible frame can arise from different cost structures, so each route chapter opens with a route-specific cost decomposition. The following summaries fix the shared vocabulary.

Throughout these decompositions, C denotes an additive work or resource quantity in one declared unit, such as FLOPs, GPU-seconds, or measured serial-stage milliseconds. Different units cannot be added. If stages overlap, their sum is not elapsed latency. For ideal pipeline stages with times $t_1, \dots, t_m$, first-output processing time is approximately $\sum_i t_i$ plus input buffering,

Table 2 Worked hybrid assignments. Each row records independent architectural properties; the last column applies the ordered editorial rule. Source editions and exact locations are in snapshot VH-2026-10-09.

Method	Appearance / personalization	Temporal state and sampling / RGB path	Primary route and decisive test
MIDAS [26]	Dense DC-AE frame latents; reference conditioned	Next-frame backbone state from committed frame tokens; four-step local flow head; DC-AE decode	AR: noise enters a separate frame-local head. Codec history and AR context are separate caches.
Body of Her [4]	Continuous video tokens; conditioned humanoid	Causal audiovisual history; current-frame tokens sampled in parallel by a compact diffusion head; video decode	AR: a fixed causal state conditions local token sampling. Deployment cache capacity is NR.
EARTalking [187]	Wan-VAE visual latents; portrait conditioned	Frame-causal backbone; spatial masked AR; local DDPM head; VAE decode	AR: frame-state update precedes 36 spatial steps and 25-step local denoising.
X-Streamer [190]	LTX-VAE chunks; portrait conditioned	Actor evaluates current noisy video tokens and timestep against video history and Thinker conditions; DDIM; VAE decode	Diffusion: the temporal video backbone is itself the iterative denoiser. The Thinker is not the visual owner.
StyleAvatar [178]	Subject-specific facial UV and background features; per-person training	3DMM/UV rendering and feature composition; adversarial StyleUNets produce non-face detail and final RGB	Render: explicit UV appearance is geometrically queried. GAN training is a secondary annotation.
LivePortrait [58]	Source feature volume; one-shot	Compact implicit keypoints; dense deformation transports source features; learned image decoder	Render: geometric transport supplies appearance despite the learned RGB generator.

whereas steady-state chunk cadence is bounded below by $\max_i t_i$; communication and contention can increase both. Enrollment is charged separately or amortized over a declared number of frames.

GAN-based synthesis. After training, a GAN generator can synthesize dense appearance without an iterative sampling loop [54, 81, 82]. One feed-forward evaluation describes call count, not absolute latency: preprocessing, network size, output resolution, compositing, and hardware still determine runtime. Route-specific speed claims therefore require a measured boundary rather than the presence of a discriminator or a single-pass architecture.

Diffusion and flow matching. Diffusion [65] and flow-matching [50, 106] models traverse a noise-to-data path through repeated function evaluations. Their cost depends on the generated unit (frame, clip, or chunk), strict number of function evaluations (NFE), cost per evaluation, conditioning passes, latent codec [88, 137, 146], and system scheduling. Step reduction [116, 147, 160], caching, and parallelism reduce different terms and should not be collapsed into a single speed label.

Autoregressive visual state. Autoregressive models factor a sequence and predict future units from prior state [13, 143, 191], over discrete tokens [171] or continuous latents [23]. Runtime depends on token or block granularity, serial depth, attention, cache updates, local decoding, and output rate. Key-value (KV) caching can avoid recomputing past projections while still grow-

ing memory; bounded windows, recurrent summaries, compression, eviction, or resets are separate Stable mechanisms. Serial depth itself can be attacked with parallel or speculative emission [93].

Render-based synthesis. Render-based systems amortize a subject-specific appearance representation and repeatedly update or query it using compact controls. Their costs include enrollment or scene construction, upstream tracking, controller evaluation, deformation or field queries, rasterization or volume integration, neural decoding, compositing, and input/output (I/O). A renderer call is not universally cheap or one-step: neural radiance field (NeRF) ray sampling and Gaussian sorting expose different cost terms [83, 126, 127]. Under our accounting rule, a fixed scene can anchor identity without proving long-session stability of the complete pipeline.

2.5 Conditions and Intermediate Representations

Runtime conditions specify what the person should do. Common signals include speech audio, text, pose, expression, gaze, and a driving video; pose and expression may themselves be estimated by upstream trackers whose cost belongs to the deployed pipeline [18, 202]. Enrollment evidence specifies who or what is rendered and may be a single reference, a short monocular recording, calibrated multiview capture, or a pre-built asset.

Intermediate representations determine the interface between control and appearance. Landmarks and keypoints are interpretable, but they omit appearance. Facial action units [49], three-dimensional morphable models (3DMMs), Faces Learned with an Articulated Model and Expressions (FLAME), the Skinned Multi-Person Linear Model (SMPL) and its expressive extension SMPL-X, and blendshape parameters expose compact geometric controls [99, 112, 133]. Flow fields and identity-conditioned warps transport source features. Learned motion latents trade semantics for capacity, whereas visual tokens or latents carry dense appearance state. Neural fields and Gaussian primitives encode subject-specific scenes [83, 126]. Dimensionality alone does not determine speed. Compact controls may feed an expensive renderer, while a dense latent may use a fast decoder.

For comparison, we record anatomical coverage (face, upper body, full body, hands), personalization regime, representation owner, decoder/renderer, and evaluation boundary. These axes remain orthogonal to the four primary routes.

2.6 Common Acceleration and Measurement Terms

Generic acceleration acts on distinct Fast terms [151]. The following categories are the shared skeleton of every route’s Fast analysis; each category still requires a measured system boundary:

- **Fewer evaluations:** distillation, consistency objectives, solver reduction, block prediction, or speculative emission reduce strict NFE or serial calls [93, 116, 160, 205].
- **Cheaper evaluations:** representation compression, sparse computation, lightweight backbones, quantization, and optimized kernels reduce the cost of each call. For attention, I/O-aware FlashAttention reorganizes exact-attention computation in 16-bit floating point (FP16) or bfloat16 (BF16) to reduce high-bandwidth-memory traffic [41, 42], whereas FlashAttention-3 also provides a Hopper 8-bit floating-point (FP8) path whose speed lever is lower-precision Tensor Core execution and whose block quantization and incoherent processing control quantization error [149]. These standalone kernel results do not establish an end-to-end virtual-human speedup.
- **Reuse:** feature, condition, reference, or KV caches reuse a precisely named object across denoising steps, frames, chunks, or autoregressive history [107, 118, 121].

- **Decoder/renderer reduction:** efficient codecs, bounded ray samples, primitive pruning, rasterizer design, and regional synthesis reduce visible-output cost.
- **System optimization:** compilation, mixed precision, memory layout, I/O, pipelining, and parallel execution change the measured pipeline boundary.

The following terms are not interchangeable. *Throughput* is completed output per wall-clock time after the disclosed operating condition is reached. *Frames per second (FPS)* is frame throughput at a stated resolution and boundary; it is not content frame rate or latency. *Real-time factor (RTF)* is wall-clock time divided by generated media duration under a stated convention, so values below one are faster than playback. *First-output latency* is the umbrella term for the interval from a declared input-arrival event to the first committed output of any type. *Time to first frame (TTFF)* is reserved for the case in which that first committed output is an observable frame. *Inter-chunk latency* measures successive committed outputs and includes the declared buffering policy. *Number of function evaluations (NFE)* counts evaluations of the iterative vector field or denoiser for the stated generated unit.

Every reported number should identify hardware, device count, resolution, precision, batch or concurrency, warm-up, measurement window, and included components. When these fields or the evaluated duration are absent from the primary source, we report them as not reported (NR) rather than infer deployment capability from architecture or title.

Matched timing quantities can be converted. If N completed frames take w seconds and the media rate is f frames/s, then $\text{FPS} = N/w$ and $\text{RTF} = wf/N = f/\text{FPS}$, provided both use the same measured interval and pipeline boundary. Neither conversion yields TTFF. For example, a 12-frame chunk generated in 0.30 s has throughput 40 FPS and RTF 0.625 at 25 FPS playback, yet collecting 0.48 s of audio before processing already imposes a different startup delay. This is an illustrative calculation, not a measured system result.

2.7 Failure Modes and Mechanism Evidence

Stability means preserving the intended attributes while allowing the requested changes. We distinguish five observable failures: *appearance drift*, including identity, color, and mouth-detail degradation; *temporal discontinuity*, including frame jitter and chunk-boundary jumps; *control failure*, including lip desynchronization and suppression of requested motion; *geometric failure*, including tracking wobble, deformation, disocclusion, and com-

positing errors; and *service failure*, including stalls, memory growth, and resets. Only some of these require recursive visual feedback. A fixed appearance asset can remain recognizable while its motion tracker jitters, and an unchanged but incorrect face can have zero measured drift.

Five recurring mechanisms help organize the streaming-generation literature. Their names describe operations, not guarantees. The evidence is strongest for streaming diffusion; applications to other representations must be checked individually.

L1: Persistent identity anchor. Retain a source image, appearance asset, or designated reference latent across updates. This supplies a stable appearance reference. A fixed anchor can also oppose intended changes; that trade-off requires a change-following test rather than an identity score alone.

L2: Bounded multiscale temporal memory. Preserve recent state and selected or compressed older temporal state under a documented capacity and update policy. We reserve L2 for this combination. A rolling KV window, a codec feature cache, and a learned detail-retrieval bank receive separate annotations. PersonaLive’s selected history has an unspecified capacity and eviction rule, so it is a selected-history mechanism rather than verified bounded L2 [102]. SoulX-LiveAct specifies fixed reference, long-term, and short-term slots and is a documented L2 example [227].

L3: Position re-indexing. Reassign temporal positions as a cache advances, for example by storing keys before rotary position embedding (RoPE) or using cyclic indices. This controls positional extrapolation; it does not restore information removed by eviction. EARTalking’s cyclic positions and Live Avatar’s Rolling RoPE instantiate different policies [76, 187].

L4: History-robust training. We distinguish $L4-G$, training on model-generated histories, from $L4-C$, corrupting ground-truth histories with noise or dropped patches. Self-Forcing exemplifies $L4-G$; MIDAS, EARTalking, and Hallo2 provide $L4-C$ examples [26, 39, 75, 187]. The former reduces a training–inference mismatch over the training rollout distribution. Neither subtype establishes distribution matching for arbitrarily long or differently conditioned deployment.

L5: Explicit boundary-state transfer. Pass a designated motion latent, overlap state, or boundary representation from one chunk to the next. JoyAvatar’s Motion Condition Injection and Avatar V’s motion propagation illustrate the operation [96, 103]. Ordinary attention to previous frames is insufficient for an L5 annotation. Transfer can improve continuity or propagate an error; its direction and magnitude are empirical questions.

Every method–mechanism association is accompanied by an evidence status. **A** denotes documented architecture; **Q**, a qualitative output example; **T**, quantitative measurements at stated temporal positions or durations; and **B**, a component ablation under the source protocol. These tags can coexist. T is not equivalent to a full continuous-session audit, and B need not be longitudinal. A proposed transfer is marked **H** (hypothesis) and does not count as a documented implementation. For example, EARTalking supplies A and B for its anchor and history corruption, with a qualitative example at three training-window lengths, but no minute-scale time-binned study [187]. Live Avatar supplies A, B, and sampled-window T under repeated audio [76]. The evidence sheet records the exact scope of each association.

This framework also preserves evidence outside L1–L5. Mouth-detail tests, pose-drift ablations, and tracking or wobble evaluations remain useful even when a method lacks a complete long-session protocol. We use them to identify which failure has been tested, rather than treating an absent joint protocol as an absence of all temporal evidence.

3 GAN-Based Methods

Adversarially trained feed-forward generators produce a frame or image region in a single generator evaluation [54, 81, 82]. This structure avoids an iterative sampling loop, but runtime still depends on the network, conditioning, and output-processing stages. We review three ways of reducing this work: regional synthesis, compact control, and reusable appearance. We then examine temporal consistency, which requires particular attention when the visual generator operates independently on each frame.

3.1 Scope, Ownership, and Cost Boundary

A method is assigned to this route when an adversarially trained feed-forward generator owns final dense or regional visual synthesis after the render-, autoregressive-, and diffusion-ownership tests in Section 2.3. A discriminator is normally used only during training, so removing it from the deployed graph is not a route test. Conversely, an adversarial loss may regularize a warper, neural field, or diffusion decoder without making that system GAN-primary. The relevant questions are what representation is generated, which module owns appearance, and how RGB is reconstructed.

The training objective and inference path must both be checked when assigning a method. Wav2Lip reports

Runtime operation	What changes when a new chunk is committed?
L1: keep a reference	Anchor a persists; recent generated state r_t advances. Changing r_t does not automatically replace a .
L2: update bounded memory	$(a, m_t, r_t) \rightarrow (a, \text{select/compress}(m_t, r_t), r_{t+1})$. Evict displaced recent state and cap both memory tiers. A recent-only window has no older tier.
L3: re-index positions	Cached content remains; its temporal indices are reassigned to the supported range. Re-indexing does not undo content errors.
L5: transfer a boundary	Chunk t exports boundary state b_t ; the next chunk initializes or conditions its motion on b_t .
Training operation	Which history enters the next prediction?
L4-G: generated history	$\hat{x}_1 = G(\epsilon_1, c_1)$; $\hat{x}_2 = G(\epsilon_2, c_2, \hat{x}_1)$; supervise the rollout. The second step sees a model error.
L4-C: corrupted history	$\tilde{x}_1 = \text{corrupt}(x_1^{GT})$; predict x_2 conditioned on $\tilde{x}_1$. Noise is sampled from an augmentation rule.

Fig. 5 State operations behind the stability vocabulary. Runtime memory updates and explicit boundary transfer are separate from the two training constructions. A memory budget bounds storage, whereas output measurements establish whether the retained state is useful.

a base generator without the visual-quality adversarial term and a Wav2Lip+GAN variant with a separate quality discriminator [138]; only the latter is included in the GAN comparison. MuseTalk reuses a pretrained variational autoencoder (VAE) and a U-Net architecture associated with latent diffusion, but performs one-call regional prediction with reconstruction, synchronization, and face/lip adversarial losses [218]. It has no diffusion objective or iterative sampler. Thus, architectural ancestry alone does not determine the route.

For a feed-forward visual path, a useful first-order deployment decomposition is

$$C_{\text{frame}} = C_{\text{condition}} + C_{\text{generator}} + C_{\text{decode/compose}} + C_{\text{system}}. \quad (3)$$

The first term covers audio, identity, pose, expression, and reference encoding. The second covers the network that predicts pixels or dense visual features. The third covers decoding, parsing, blending, refinement, and super-resolution. The final term covers data movement, precision conversion, scheduling, and I/O. The methods below each attack a different term; none establishes time to first frame (TTFF) or steady-state throughput unless the source defines the timer and system boundary.

3.2 Regional Synthesis: Shrinking $C_{\text{generator}}$

The most direct saving is to restrict synthesis to the region that actually changes. Wav2Lip conditions a feed-forward face decoder on identity, a lower-half-masked pose prior, and speech, and generates a dense RGB face crop frame by frame; a frozen five-frame synchronization expert supplies a training loss rather than recurrent inference state [138]. Under the paper’s LRS2-derived protocol, the base and GAN variants expose a small synchronization–fidelity trade-off measured by

lip-sync error distance (LSE-D), lip-sync error confidence (LSE-C), and Fréchet Inception Distance (FID): 6.386/7.789/4.887 versus 6.469/7.781/4.446. The source reports no runtime measurements, but the design pattern of synthesizing the mouth and keeping the canvas became the template for a large lip-editing family, including VideoReTalking’s editing pipeline [27], DInet’s deformation-inpainting for high-resolution dubbing [219], landmark- and appearance-prior variants [230], memory-augmented mouth retrieval [132, 165], and style-space lip synchronization [57, 86], with ReSyncer unifying style-based synchronization across tasks [56].

MuseTalk moves regional synthesis into a pretrained VAE latent space, which shrinks the generated tensor a second time: its multimodal U-Net predicts a lower-face latent in one call, and a precomputed mask is reused for compositing [218]. The authors report 30 FPS for 256×256 output on one V100 with preloaded data (component inclusion NR). The supplement identifies occasional jitter from single-frame generation, the canonical Stable weakness of stateless regional synthesis, taken up in Section 3.6.

Training-only critics can improve a regional generator without adding a deployed inference module. TalkLip uses a frozen AV-HuBERT-derived lip-reading expert and reports improved word intelligibility over Wav2Lip on its LRW/LRS2 protocols [175]. Synchronization experts, lip-reading experts, and quality discriminators therefore offer additional supervision without a corresponding critic evaluation at runtime. Their training cost and the generator’s own inference cost remain separate considerations.

3.3 Structured Control and Codebooks: Shrinking $C_{\text{condition}}$ and Bounding the Output Space

A second family separates identity from compact motion controls. MakeItTalk splits content and speaker-aware landmark motion [233], while PC-AVS separates pose, content, and identity in latent space [231]. EDTalk factorizes mouth, pose, and expression into orthogonal basis banks; its shared adversarial image generator produces RGB, and a normalizing flow samples compact pose-bank weights [163]. Emotional adaptation provides another use of disentangled control [53].

StyleGAN-based reenactment instead controls a pre-trained generator by editing latent directions, masking style channels, or refining inversions [10–12, 20]. Variants address continuous emotion [6], semantic expression [131], cross-domain heads [80], and text-guided editing [16]. These methods make the control interface compact, but do not necessarily make control extraction inexpensive: inversion and feature estimation can add work, and the full image generator remains on the inference path.

Codebooks represent generated appearance through a learned set of reusable features. FlowVQTalker combines autoregressive normalizing-flow controllers for compact 3DMM expression and pose coefficients with an adversarially trained vector-quantized image generator, which decodes a spatial code sequence into a 512×512 face [164]. A grid-based codebook provides a related efficiency-oriented representation for person-specific reenactment [177].

Retrieval reuses detail compatible with the source identity. FixTalk compresses fourth-layer detail features from $512 \times 32 \times 32$ to a 1×512 token and retrieves aligned detail before generation; its Enhanced Motion Indicator suppresses identity leakage from driving features [162]. The reported throughput is 27.601 FPS on an A100 (80 GB) at 512×512 , with complete timer composition NR. This supports the reported operating point, but does not independently quantify the speed contribution of retrieval or compression.

3.4 Personalized Compositional Generators: Amortizing Appearance

Appearance reuse can be made stronger through subject-specific training, trading enrollment cost for per-frame savings. Adversarial synthesis can also be combined with geometric appearance reuse; the primary route then follows the first matching rule. Live Speech Portraits is the route’s early real-time landmark: a person-specific pipeline from streaming audio through predicted motion to adversarial image synthesis, reported at 30 FPS

with over 27 FPS attributed to generation on a consumer GPU [113]. StyleAvatar is a contextual render-GAN hybrid, counted in Table 10 under the consistent render-first rule. It scales the idea to a compositional generator: two StyleGAN modules for static facial-UV and background features, a StyleUNet for non-facial foreground, Neural Textures with a learned mask for regional composition, and a second StyleUNet for final RGB [178]. With two 16-bit TensorRT StyleUNets, the authors report a source-labeled “live system” rate of 35 FPS (28 ms/frame) on one RTX 3090, with 20 ms for the model-only GPU boundary. The cost structure is instructive: static regional features are computed once (enrollment, reported at roughly two to three hours of person-specific training plus about four more for teeth convergence), so the per-frame path touches only the dynamic StyleUNets, a direct reduction of $C_{\text{generator}}$ purchased with C_{enroll} . RAM-Avatar extends person-specific adversarial synthesis to full-body control with real-time intent [45].

3.5 Fast Design Guidance

Table 3 consolidates the evidence. Four levers recur, each acting on a different term of Equation (3):

1. **Synthesize only what changes** ($C_{\text{generator}} \downarrow$). Face-crop or lower-face prediction reduces the generated area; latent-space prediction, as in MuseTalk, further compresses the representation. The saving depends on crop size and codec compression. Compositing and boundary blending remain necessary, with seams and mouth interiors requiring separate quality evaluation.
2. **Drive with compact controls** ($C_{\text{condition}} \downarrow$). Basis banks, 3DMM coefficients, latent directions, and codebook indices replace dense driving features. Cost: expressiveness is bounded by the control space; motion outside it cannot be generated.
3. **Reuse appearance features** ($C_{\text{generator}} \downarrow$). Person-specific static feature banks, as in StyleAvatar, trade enrollment for repeated savings. Reference-conditioned retrieval, as in FixTalk, can instead support one-shot operation. Their preparation costs and generalization requirements should therefore be reported separately.
4. **Optimize the fixed path** ($C_{\text{system}} \downarrow$). Compilation, reduced precision, and kernel fusion can reduce deployment overhead. StyleAvatar provides an FP16 TensorRT example. The gains depend on the hardware and implementation; its model-only 20 ms remains distinct from system latency.

These mechanisms address different components and can be combined, although their total benefit depends on the remaining pipeline. Face detection, alignment, audio encoding, and I/O may become substantial once generation is inexpensive. The FPS values in Tables 3 and 10 use different resolutions, devices, personalization regimes, and timing boundaries, so they describe individual systems rather than a controlled ranking.

3.6 Stable Design Guidance

Framewise generation avoids feedback of a previously synthesized frame when that frame is not reused as an input. This removes one source of recursive error, but does not enforce temporal consistency. MuseTalk reports single-frame jitter; variation in pose estimates or reference features can also produce inconsistent output. Recurrent controllers and retrieval memories introduce additional state whose errors must be evaluated independently of the image generator.

The drift-control patterns in Section 2.7 help distinguish these cases. A fixed reference or person-specific feature bank serves as an identity anchor (L1). FixTalk’s retrieval memory and FlowVQTalker’s coefficient history supply persistent information, although they do not by themselves establish bounded multiscale temporal memory (L2). Position re-indexing (L3) is relevant only where a sequential component uses positions. Evidence for self-history training (L4) and explicit motion inheritance (L5) is limited in the systems reviewed here. None of the papers in Table 3 reports a joint minute-scale, time-binned evaluation of identity, synchronization, motion, and visual degradation. The long-session benefit of avoiding visual feedback therefore remains a hypothesis to test.

3.7 Synthesis and Open Questions

Regional methods require evaluation that separates synthesized content from preserved source pixels, including artifacts at their boundary. Compact controls should be assessed for expressive coverage and extraction cost, while personalized generators require enrollment reporting alongside runtime. For stability, a duration-binned comparison could test whether avoiding visual feedback reduces drift and how much framewise jitter remains. The reviewed sources do not provide that joint comparison. Reporting the generated region, resolution, personalization, preparation cost, and timing boundary would make these trade-offs easier to assess.

GAN-based methods reduce repeated synthesis through feed-forward prediction, regional output, com-

pact control, and reusable appearance. Their temporal behavior depends on both the image generator and its conditioning components. The reviewed sources demonstrate several efficiency mechanisms, but their heterogeneous timing boundaries and limited longitudinal evaluation leave the joint efficiency–stability trade-off unresolved.

4 Diffusion-Based Methods

Diffusion and flow models support generation conditioned on identity, speech, text, pose, and previous video [65, 68, 105, 106, 167]. Their iterative inference motivates methods that reduce sampling calls, compress the generated representation, or reuse computation. Extending generation across chunks introduces a second challenge: the model must maintain consistency while consuming its own outputs. We first review the main acceleration mechanisms, then discuss streaming generation and five recurring approaches to drift control. Architecture alone does not determine their speed relative to the other routes.

4.1 Scope and Cost Model

The reverse chain of a conventional denoising diffusion probabilistic model (DDPM) serially invokes the denoising network many times [65]. Flow matching avoids simulating an ordinary differential equation (ODE) while constructing its training targets and can learn straighter probability paths, but generation still requires integrating the learned vector field [50, 106]. With a diffusion Transformer (DiT) in video-latent space, each invocation is itself governed by the number of tokens, model width and depth, and the temporal and spatial extent of attention [134, 201]. A useful first-order decomposition of deployed generation cost is

$$C_{\text{gen}} = N_{\text{FE}}C_{\text{denoise}} + C_{\text{condition}} + C_{\text{decode/render}} + C_{\text{system}}. \quad (4)$$

Here, the first term counts the number of function evaluations (NFE) and depends on sampling steps, classifier-free guidance (CFG) branches, solver stages, and any extra network calls used to refresh a cache. The second covers speech, text, reference-image, and motion encoders; the third includes VAE decoding, super-resolution, and any separate neural renderer; and the last includes kernel scheduling, device communication, data transfer, playback buffering, and I/O. Equation (4) explains why few-step sampling does not by itself establish end-to-end speed. Once N_{FE} is small, the bottleneck

Table 3 Representative GAN-based visual generators. Timing values retain the source paper’s hardware, resolution, and component boundary and are not directly comparable. NR denotes not reported in the primary paper.

Method	Visual owner / scope	Primary Fast mechanism	Source timing boundary	Stable boundary
Wav2Lip+GAN [138]	Feed-forward RGB face-crop generator; identity-general	Face-crop rather than full-canvas synthesis; training-only sync and quality critics	FPS, hardware, resolution, TTFF, and timer NR	Stateless framewise generation; evaluated duration NR
MuseTalk [218]	U-Net-generated lower-face VAE latent; no subject adaptation	One-call regional latent prediction; reusable blend mask	30 FPS, 256 ² , one V100, preloaded data; included stages NR	Single-frame jitter reported; quantitative long-horizon evidence NR
TalkLip [175]	Feed-forward 96 ² RGB face-crop generator; identity-general	Training-only lip-reading expert improves intelligibility	FPS, latency, hardware, parameters, and timer NR	Stateless framewise generation; long-horizon evidence NR
Live Speech Portraits [113]	Person-specific adversarial portrait generator	Compact predicted motion; person-specific appearance	Source-reported 30 FPS system rate on a consumer GPU; timer boundary NR	Recurrent motion prediction; long-horizon evidence NR
EDTalk [163]	Adversarial RGB generator; mouth, pose, and expression banks; one-shot	Compact reusable bases and progressive freezing	Inference timing NR; about 7 h disentanglement training on two 3090 GPUs	Compact controller state only; long-horizon evidence NR
FlowVQ Talker [164]	Adversarial VQ image generator; full 512 ² portrait	Compact coefficient control and learned spatial codebook	Inference FPS, latency, hardware, memory, and timer NR	Coefficient history in the controller; long-horizon evidence NR
FixTalk [162]	Adversarial dense portrait generator; one-shot	Identity-suppressing EMI and compressed memory-retrieved EDI detail	27.601 FPS on A100 80 GB; 512 ² data, full timer boundary NR	Learned detail retrieval (not temporal L2); TTFF and long-horizon evidence NR

EDI: Enhanced Detail Indicator; EMI: Enhanced Motion Indicator. NR retains the source-specific extraction boundary; verification status is recorded in the supplement.

Table 4 GAN-route guidance. D marks a documented mechanism; H marks a deployment hypothesis or cross-route transfer. D does not imply that the target is met under every configuration.

Target	Design options	Trade-offs to check
Low per-frame latency, general identity	D: regional synthesis; one-call generator	Seams, mouth-interior quality; preprocessing untimed
High throughput at high resolution	D: compact controls and retrieval; H: transfer compiled FP16 serving	Control space bounds expressiveness; hardware-specific
Repeated use of one subject	D: person-specific appearance reuse	Hours-scale enrollment; no zero-shot use
Minute-scale stability	D: reference and temporal control; H: long-session deployment	Framewise jitter and controller drift; joint long-horizon evaluation NR in the reviewed systems

can move to a single large DiT call, the codec, condition encoding, or communication. Component-level gains from caching, quantization, and parallelism therefore cannot be multiplied mechanically into a system-level speedup.

For deployed systems, two service clocks matter beyond raw throughput. Time to first frame (TTFF) controls when playback can begin; it includes input acquisition, condition encoding, first-chunk denoising, first decode, and buffering, so audio buffering imposes a TTFF floor that no denoiser acceleration can remove. Steady-state inter-chunk latency and jitter control whether playback continues without stalls. A nominal playback rate and a real-time factor are neither of these. We project *Fast* onto NFE, per-call computation, decoding, and system overhead, with TTFF and steady-state throughput as its service observables; we project *Stable* onto bounded state and identity or appearance drift under self-generated history. Reported speeds retain the source timing boundary.

Dense visual diffusion generates pixels or video latents through a noise-conditioned visual backbone; a codec then reconstructs RGB. FADA, RAP, StreamAvatar, SoulX-FlashTalk, and Avatar V illustrate this path [48, 103, 152, 161, 229]. Compact motion genera-

tors followed by identity-conditioned warping or scene rendering belong at the render boundary, including IF-MDM, INFP, AudioRTA, Avatar Forcing, EmbodiedHead, FloodDiffusion, and OmniTalker [15, 85, 91, 186, 199, 217, 236]. Table 2 resolves the harder distinction between an iterative temporal backbone and a fixed causal visual state with a local iterative head.

The terminology around “flow” needs one clarification. Normalizing flows use invertible transformations with tractable Jacobians; their connection to diffusion lies at the level of probability-density evolution [130]. In current virtual-human systems, flow matching or rectified flow means regressing a velocity field from noise toward data and sampling it with an ODE solver or a distilled approximation; “simulation-free” characterizes the training objective, not one-call inference [106]. Sampling steps convert to strict NFE only when solver stages, CFG branches, and auxiliary cache calls are counted.

Across these architectures, a stable task prior organizes many of the mechanisms below: *appearance changes slowly while the driving signal changes quickly*. Identity, clothing, background, and reference style are low-frequency state that can be compressed, cached, or kept as persistent anchors. Phonemes, mouth shape, transient expressions, and gesture must track the current input. READ aligns a temporally compressed visual latent with a SpeechAE [173]; RAP introduces hybrid audio attention to compensate for strongly compressed video tokens [48]; SyncCache reuses heavy DiT residuals within one reverse trajectory while recomputing light audio blocks at every step [117]. The same asymmetry motivates long-video systems to preserve a reference sink and recent motion while compressing or evicting older history.

The main acceleration mechanisms can now be located in Equation (4). Few-step distillation, consistency training, and distribution matching reduce N_{FE} . Compressed latents, smaller backbones, sparse attention, and low-precision execution reduce C_{denoise} . Pre-encoding and cross-step, cross-block, or cross-modal caches remove repeated conditioning and attention work. Lightweight or causal codecs reduce $C_{\text{decode/render}}$. Fixed windows, fused kernels, and asynchronous pipelines reorganize C_{system} . Deployed systems usually combine several levers, and meaningful evaluation must identify the active levers and track where the bottleneck moves, rather than assigning the joint system result to a single module [76, 117, 229].

Figure 6 separates component optimization from runtime state updates. The mechanism vocabulary in Section 2.7 makes it possible to ask which failure each state operation addresses.

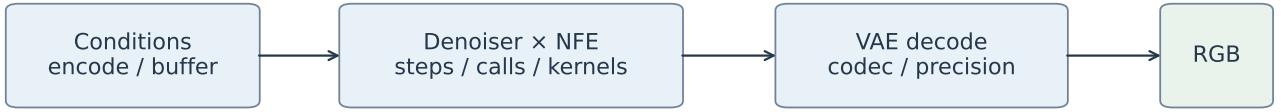
4.2 Fast I: Compact Representations and Efficient Backbones

Representation-level acceleration follows two fundamentally different paths. One diffuses in a low-dimensional motion, expression, or parametric space and delegates appearance to a separate decoder or renderer. Its per-step state is small and easy to retain, but its expressiveness is bounded by the intermediate representation and often by an identity-specific renderer. The other continues to generate dense visual content, but applies a highly compressed spatiotemporal VAE so that the DiT processes far fewer video tokens. It preserves a more open appearance model, while paying for audio-visual alignment, recovery of local high frequencies, and VAE decoding. FPS measured for these two paths does not denote the same computation.

Compact state can move the expensive stochastic process out of pixel space. IF-MDM diffuses in a 20-dimensional implicit facial-motion space before an RGB generator synthesizes the portrait. INFP uses a conditional DiT with only four blocks to generate speaking and listening motion latents (no inference timing reported). AudioRTA distills a 40-step DDIM teacher into one-step expression-latent prediction and renders through the mesh/3D-Gaussian decoder of URAvatar. Avatar Forcing applies block-causal diffusion forcing to a 512-dimensional motion latent per frame and predicts reactive dyadic behavior from both parties’ signals. EmbodiedHead conditions on the per-frame listening/speaking state, predicts a 115-dimensional FLAME motion vector with a rectified-flow DiT, and passes it to GAGAvatar [85, 91, 199, 217, 236]. FloodDiffusion studies streaming, time-varying text conditions in skeleton-motion space [15]. These methods show why compact state is efficient: the diffusion process runs over tens of dimensions rather than dense latents. Their motion-module or renderer-inclusive speeds should nevertheless not be compared directly with open-domain dense-video pipelines, and their appearance ceiling is set by the downstream renderer (Section 6).

The dense path reduces cost by compressing space and time. Stable Video Diffusion shows that latent-video quality depends not only on the denoiser, but also on image pretraining, video-data curation, and temporal-layer training [9]. HunyuanVideo, Wan, and CogVideoX organize large video generation around a spatiotemporal VAE, DiT, and text encoder [90, 172, 201]; they expose the shared computational substrate: VAE compression determines token volume, full spatiotemporal attention determines much of the per-call cost, and chunked decoding determines whether frames can be emitted before all latents are available. LTX-Video co-

A Runtime path: optimize a named stage



B What remains after step reduction?

Fewer denoiser calls leave encoding, decoding and transfers on the path.

A decoder change can alter speed and facial detail: see the PersonaLive ablation.

C Runtime history update (separate from training feedback)

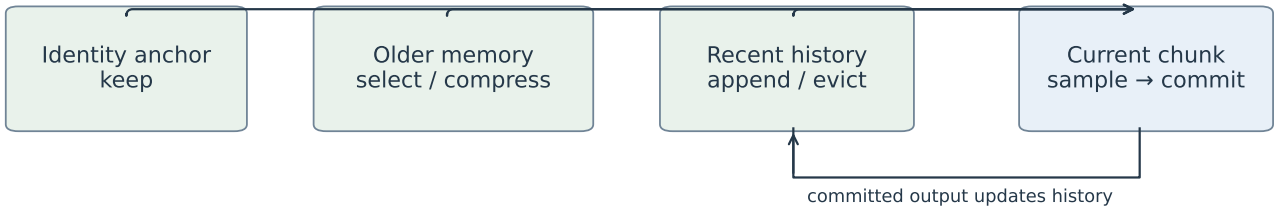


Fig. 6 Diffusion generation as a runtime path and a state-update process. (A) Acceleration acts on distinct components. (B) After denoiser-call reduction, other stages still contribute to runtime; Section 7 gives a measured decoder trade-off. (C) The committed chunk updates recent history, while older state is selected or compressed under the declared policy. The feedback path lies entirely within the diagram. Training constructions are shown separately in Figure 5.

designs a high-compression VAE with a token-matched DiT, illustrating that reducing token count can change the computation more substantially than shrinking the network [61]. For virtual humans, however, aggressive compression is especially risky around teeth, lip boundaries, eyelids, and rapid expressions.

If temporal and spatial compression factors are r_t and r_s , the video-token count scales approximately as $THW/(r_t r_s^2)$ [61, 134]. Global self-attention is quadratic in this count, and audio cross-attention depends on the number of audio tokens. Greater compression reduces attention, MLP computation, KV storage, and communication. It also increases the phonetic changes represented by each temporal token and allocates fewer spatial tokens to small facial structures. Compression must therefore be co-designed with audio downsampling, detail recovery, and decoder throughput. READ and RAP illustrate two solutions: READ uses a Temporal VAE to extend each visual token over a spatiotemporal block and trains a SpeechAE at the corresponding time scale (eight solver steps; the reported 4.421 s is backbone time for one 4.84 s, 121-frame sample) [173]; RAP combines an LTX-VAE latent, hybrid audio attention, and latent inheritance, reaching 42.41 and 39.87 FPS on one A800 at 512×512 under the authors’ protocol [48]. Both show that once the visual timeline is compressed,

frame-level speech features cannot simply be injected without accounting for how several phonetic events map to one visual token.

Dense generation can also reduce the cost of a long reference. Avatar V lets generated tokens attend to a long reference video through asymmetric sparse reference attention; reference K/V projections and their rotary position embedding (RoPE) results are computed in the first denoising step and reused for the remaining 23, followed by a one-step adversarial super-resolution stage. The paper reports an approximately threefold latency reduction relative to its unoptimized implementation, with no absolute latency or FPS [103]. The case shows that long-reference tokens can become a second attention bottleneck and that cross-step conditioning caches matter even after denoising is shortened.

After DiT distillation, VAE cost remains on the inference path (Figure 6B). PersonaLive offers TinyVAE for higher decoder throughput, but records degradation in identity, motion, and temporal metrics; MotionStream’s highest reported 29.5 FPS configuration on one H100 also uses TinyVAE [102, 153]. Wan’s causal VAE can encode and decode arbitrarily long video in chunks while caching adjacent codec features [172]. Compact representations should be evaluated with reconstruction quality, decoder latency, mouth detail, identity, synchronization,

and the actual end-to-end timing boundary, not only compression ratio or denoiser floating-point operations (FLOPs).

Conditioning cost and expression control. Audio conditioning is not a single objective. EmotiveTalk separates audio information relevant to lip motion from information associated with emotion and expression, then uses emotional video diffusion to expand the temporal and expressive signal. The design makes explicit a distinction that matters for both Fast and Stable evaluation: phonetic accuracy, motion naturalness, and identity fidelity are separate targets, and an audio encoder or diffusion sampler may improve one while leaving the others unchanged. Conditioning-side disentanglement therefore belongs in the diffusion cost ledger, including its extra encoders, attention paths, and temporal expansion, rather than being counted as a free quality module [174]. The same principle motivates emotion-specialized or mixture-of-expert controllers, but routing complexity and expert capacity must be reported separately from the visual generator.

4.3 Fast II: Few-Step and One-Step Distillation

Few-step methods shorten the reverse trajectory, but at least three operations are often conflated. Trajectory rectification or consistency training makes distant noise states easier to connect; distribution matching distillation (DMD) trains a short-path generator by contrasting real- and generated-distribution scores; and CFG distillation folds conditional and unconditional branches within a step into one student call. These mechanisms alter the path, student distribution, and calls per step, respectively, and should not all be called one-step generation. In a video DiT, a sampling step may invoke several CFG branches, while a streaming method may add a clean-latent call solely to update KV. Strict NFE must count actual inference-time network evaluations.

The generic lineage requires precise objectives. Consistency Models map points on the same probability-flow ODE trajectory to a common output; consistency distillation depends on a pretrained diffusion trajectory, whereas standalone consistency training learns directly from data [160]. Latent consistency models and their LoRA form made few-step sampling broadly practical for latent image diffusion [116]. DMD estimates a Kullback–Leibler gradient direction from the difference between real-data and generated-distribution scores and adds a perceptual regression term anchoring the student to its teacher’s conditional sample [205]; DMD2 addresses stale generated-score estimates and unstable optimization through two-timescale updates, adversarial

supervision on real data, and backward simulation of inference-relevant states [204]. These generic image results motivate virtual-human students, but do not settle identity, synchronization, or self-history errors.

A reproducible call ledger should separate the main denoiser, auxiliary score networks used only during training, and the decoder. For each inference step, the main-network count is the product or sum induced by CFG branches, solver substages, and extra cache-refresh calls. Teacher and critic evaluations during training do not belong in deployed NFE, whereas an inference-time clean-latent forward does. If a paper reports steps but not solver or guidance calls, the survey retains “steps” rather than inventing NFE; if it reports an NFE ratio without wall-clock measurements on matched hardware, the ratio is not rewritten as a system speedup.

FADA provides a relatively explicit ledger. Its PeR-Flow student reduces a 25-step DDIM teacher to six denoising steps. Because the teacher executes three multi-CFG branches per step, the authors count 75 denoising-network evaluations (NFE-D). The Balanced student retains the three branches and uses 18 NFE-D, whereas the Fast student replaces them with a learnable CFG token and uses 6 NFE-D. The corresponding $4.17\times$ and $12.5\times$ reductions refer to denoiser calls; the paper reports neither inference hardware nor wall-clock time [229]. Trajectory shortening and branch folding are thus distinct levers: a six-step model can still repeat a large network three times at every step.

Live Avatar illustrates why cache calls also belong in this accounting. Its undistilled baseline uses 80 NFE. DMD removes CFG and yields four denoising steps, but a conventional Self-Forcing implementation adds one clean-cache refresh and therefore totals five NFE. History Corrupt instead keeps historical KV at the same noise level as the current block, eliminating that clean refresh and reducing the deployed ledger to exactly four NFE [76]. The mechanism simultaneously avoids one call and produces local, step-matched noisy KV for a pipeline that assigns one timestep to each device.

LLIA and TurboTalk represent direct few-step training and progressive one-step distillation. LLIA trains two- and four-step students with consistency and adversarial objectives, then combines them with TensorRT, 8-bit integer (INT8) execution, and VAE–U-Net overlap; at 384×384 on an RTX 4090D, 78 FPS is measured for a 12-frame steady-state chunk, with a source-defined approximately 140-ms initial/state-switch response using a four-frame startup segment [208]. TurboTalk first obtains a four-step DMD student and progressively distills three-, two-, and one-step students; its claimed $120\times$ ratio compares approximately 120 NFE per InfiniteTalk chunk with the final author-counted 1 NFE per chunk,

with no wall-clock result [109]. NFE ratios are valuable algorithmic evidence, but they do not substitute for deployment timing.

Block-autoregressive distillation must additionally close the gap between ground-truth histories used in training and self-generated histories seen at deployment. This is a Stable concern (pattern L4-G in Section 2.7) that surfaces inside the distillation objective. MotionStream’s Self Forcing, StreamAvatar’s and LiveTalk’s on-policy rollouts, SoulX-FlashTalk’s multi-block retrospective self-correction, and AsymTalker’s asymmetric distillation all expose the student to some of its own errors [28, 114, 152, 153, 161]. SoulX-FlashTalk trains a four-NFE student over rollouts of up to five blocks and reports 32 FPS with a 0.87 s startup time at 720×416 on eight H800 GPUs [152]. AsymTalker gives the teacher a ground-truth continuity reference and the student a self-generated one, then reports FID and ArcFace curves through 600 s [114]. A streaming student must match not only an isolated teacher clip, but also the state distribution induced by its own rollout.

Virtual-human-specific rewards or timestep schedules can protect signals that generic DMD discards. Lip Forcing observes that teacher CFG affects fidelity and lip synchronization differently along the trajectory; it activates CFG only in an intermediate teacher-score window and weights DMD updates with SyncNet confidence. The deployed student performs two denoising NFE without CFG plus one clean-latent cache forward; its 1.3B and 14B variants report 31.58 and 15.11 FPS on one H100 at 512×512 , excluding speech preprocessing, detection, and paste-back [30]. Hallo-Live uses VideoAlign, AudioBox, and a SyncNet proxy in preference-based DMD and reports joint audio–video throughput of 20.38 FPS with 0.94 s reported latency on two H200 GPUs [95]. Such rewards may preserve synchronization or identity while inheriting evaluator bias. A convincing comparison should report sampling steps, CFG branches, actual NFE, matched-hardware wall-clock time, teacher–student quality gaps, and long-horizon degradation under self-generated history.

4.4 Fast III: Computation Reuse and System Parallelism

When several denoising calls remain, adjacent steps contain exploitable redundancy. LightningCP runs the full U-Net at selected key timesteps, reuses high-level decoder features at the others, and assigns finer attention to the foreground; on its Hallo setup, fixed-clip U-Net latency falls from 23.692 to 7.528 s (3.15 $\times$), though the baseline uses one A800 whereas the proposed system

uses an unspecified number of A800 GPUs [111]. Follow-Your-Emoji-Faster uses a timestep-aware information cache under the same 30-step DDIM schedule, reducing U-Net denoising for 64 frames at 256×256 from 6.28 to 2.36 s (2.66 $\times$; hardware NR) [120]. Both results are fixed-clip denoiser acceleration rather than FPS. The same lever generalizes: flow caching skips recomputable segments of the velocity field in autoregressive video generation [121], and speculative decoding, which drafts cheap candidate continuations verified by the full model, has been transplanted from language models [93] to autoregressive video diffusion [71].

The slow-appearance/fast-speech asymmetry permits spatially and modally selective reuse. SyncCache’s spatial-aware probe applies a soft portrait mask to the difference between outputs of the first Transformer block at adjacent denoising steps and accumulates that error; only a threshold crossing triggers a full forward and refresh. Its modality-dynamic cache operates within one reverse trajectory: after a full-computation step, selected heavy DiT inter-block residuals are reused at later steps, while lightweight audio blocks are recomputed at every step. On eight A800 GPUs, the authors report a HunyuanVideo-Avatar path decreasing from 524 to 127 s (4.12 $\times$) and a Wan path from 113 to 30.15 s (3.75 $\times$) [117].

Two cache axes are orthogonal, and matching each cache to the update frequency of the state it stores is the route’s general reuse rule. *Denoising-time reuse* shares features between sampling steps of the same output chunk; static references, background, and slowly varying high-level features are plausible candidates, but feature distributions change from the high-noise to detail-refinement regime. *Sequence-time reuse* carries KV or clean latents from one output chunk to the next; it enables continuation, but can propagate a mouth, pose, or texture error. Spatially, the background tolerates less frequent refresh than the mouth; across modalities, identity references tolerate longer reuse than current speech. Avatar V caches reference K/V that is invariant across denoising steps; SoulX-LiveAct’s Neighbor Forcing aligns historical and current chunks at the same diffusion step so that existing KV can be reused across chunks, while ConvKV compresses older history [103, 227]. Hybrid KV compression schemes extend the same idea for autoregressive video diffusion [79]. Cache evaluations should expose key-step selection, refresh thresholds, hit rates, warm-up cost, peak memory, and quality as cache depth increases [76, 117, 227].

System design determines whether algorithmic savings become a stable output pipeline. LLIA overlaps VAE and U-Net execution under TensorRT and INT8. StreamAvatar overlaps DiT and VAE on two H800

Table 5 Compact Fast evidence for diffusion systems. Values retain source units and boundaries. Full submitted notes and newly checked configurations are separated in the evidence supplement. NFE-D counts denoiser calls; NR is source-specific, C an unresolved conflict.

Method	Sampling / call evidence	Reported timing	Resource and boundary
FADA [229]	6 steps; 75 $\rightarrow$ 18/6 NFE-D	4.17/12.5 $\times$ call reduction	Wall-clock time and hardware NR
LLIA [208]	2/4 steps; strict NFE NR	78 FPS; $\sim$ 140 ms response	RTX 4090D, 384 ² ; different startup/steady chunks
RAP [48]	Strict NFE NR	42.41 / 39.87 FPS	One A800, 512 ² ; full boundary NR
LightningCP [111]	Steps/NFE NR	U-Net: 23.692 $\rightarrow$ 7.528 s/clip	One-GPU baseline; multi-A800 method
Follow-Your-Emoji-Faster [120]	30 DDIM steps	U-Net: 6.28 $\rightarrow$ 2.36 s	64 frames, 256 ² ; device NR
PersonaLive [102]	4 steps	15.82 FPS; 0.253 s/chunk	One H100, 512 ² ; default decoder
MotionStream [153]	3 NFE/chunk	29.5 FPS; 0.39 s latency	One H100, 480p, TinyVAE; general motion
StreamAvatar [161]	3 steps; strict NFE NR	$\sim$ 1.20 s first output	Two H800, 928 $\times$ 704; 25 FPS is media rate
Hallo-Live [95]	Steps/NFE NR	20.38 FPS; 0.94 s latency	Two H200; full boundary NR
Lip Forcing [30]	2 denoising + 1 cache call	31.58 / 15.11 FPS	One H100, 512 ² ; speech preprocessing excluded
FlowAct-R1 [179]	3 base NFE; refiner NR	25 FPS; $\sim$ 1.5 s TTFF	A100 platform, 480p; device count/full boundary NR
Live Avatar [76]	4 NFE	45.2 FPS; 1.21 s TTFF	Five H100, 720 $\times$ 400, v6; network excluded
SoulX-FlashTalk [152]	4 NFE	32 FPS; 0.87 s startup	Eight H800, 720 $\times$ 416
SoulX-FlashHead [210]	Steps/NFE NR	Lite: 96; Pro: 10.81 FPS	C: H20 versus RTX 4090 for Lite
SoulX-LiveAct [227]	3 steps; strict NFE NR	20 FPS; 0.94 s latency	Two H100/H200; latency definition NR
Avatar V [103]	24 base steps + 1 SR step	$\sim$ 3 $\times$ relative latency gain	Eight unspecified GPUs; absolute timing NR

GPUs; its approximately 1.20 s time to first output decomposes into 0.48 s input buffering, 0.33 s for the first DiT output, and 0.39 s for the first VAE output, and its 25 FPS is the target playback rate rather than measured full-pipeline throughput [161]. FlowAct-R1 combines a three-NFE base streaming cycle without CFG, selective FP8, frame-level hybrid parallelism, kernel fusion, and asynchronous DiT/VAE execution, reporting 25 FPS and about 1.5 s TTFF at 480p on an “NVIDIA A100 platform” (GPU count NR) [179]. SoulX-FlashTalk assigns pipeline stages to speech processing, four DiT steps, VAE, and history encoding [152].

Live Avatar’s Timestep-forcing Pipeline Parallelism (TPP) partitions denoising time rather than network layers. In the audited v6 edition, four H100 GPUs each execute one DiT timestep with step-matched rolling KV; a fifth H100 decodes with a causal VAE. Section 7 analyzes the measured serving configurations. Table 3 reports 45.2 FPS and 1.21 s TTFF excluding network transfer, while Appendix C.4, Table 10 reports 0.94 s for the stated default three-latent block. We preserve this unresolved configuration discrepancy rather than substituting one value for the other [76]. Pipeline throughput depends on the slowest active stage, whereas startup includes the first traversal and buffering; neither follows from device count alone.

Table 5 makes the comparison problem explicit. Motion-parameter prediction is lighter than dense-video synthesis, TinyVAE and device count can alter the throughput of the same student, and NFE ratios exclude

decoders and I/O. The useful object of comparison is therefore a quality–speed Pareto curve under a fixed task, frame size, hardware budget, and timing boundary, with algorithmic call reduction, component reuse, and system parallelism reported separately.

4.5 Streaming Generation: Where Fast Meets Stable

Extending a clip generator to continuous output is the point where the route’s Fast and Stable problems become one design problem. Chunkwise emission bounds per-chunk latency (Fast), but it forces the model to condition on its own committed history (Stable), and the chunk size trades the two directly: smaller chunks reduce buffering delay and TTFF while increasing scheduling, cache-update, and decoder-startup overhead; larger chunks improve parallel efficiency while raising TTFF. Equal FPS can therefore coexist with very different service behavior, which is why TTFF and inter-chunk latency are separate columns in our tables.

Diffusion Forcing provides the mechanism-level starting point. It independently samples a noise level for every token, unifying prediction, filtering, and imputation under a causal sequence model; at inference, a configurable two-dimensional noise schedule controls the denoising progress at each position [22]. Self-Forcing changes the training distribution: the student consumes its own generated history during an on-policy rollout and matches the distribution of the full autoregressive trajectory while maintaining a rolling KV cache [75].

This addresses the discrepancy between ground-truth history in training and self-history at deployment (L4-G in Section 2.7) and is not itself an NFE-reduction method. CausVid distills a bidirectional video-diffusion teacher into a student that generates chunks in order, using ODE initialization, asymmetric DMD, and rolling KV (9.4 FPS with four sampling steps on one H100, generic video) [206]. Causal Forcing first constructs an autoregressive teacher aligned with the deployed architecture and then applies asymmetric DMD with self-rollout (17.0 FPS on one H100, generic video) [235]. Rolling Forcing pushes the same lineage to real-time long-video generation with rolling-window joint denoising [108]. Foundation-model streaming variants now package these mechanisms at scale: Wan-Streamer targets end-to-end real-time interactive generation, with its second version raising resolution at unchanged latency [72, 73], and Vidu S1 makes the same real-time streaming claim for a production system [213].

Virtual-human systems add task state to this lineage. PersonaLive advances micro-chunks through four noise levels [102]. StreamAvatar and LiveTalk distill a bidirectional teacher into a chunkwise student and add reference or identity sinks plus self-generated history [28, 161]. EchoTorrent generates chunks in order with a reference sink and tail-only forcing [124]. SoulX-FlashTalk uses multi-block self-correction; SoulX-LiveAct maintains fixed multiscale KV; Lip Forcing introduces synchronization-aware distillation with an explicit clean-cache call; and FlowAct-R1 extends chunkwise generation to full-body behavior with MLLM action planning [30, 152, 179, 227]. Sliding-window systems organize the same state for offline long inputs: StableAvatar’s Dynamic Weighted Sliding Window (2.32 min per 81 frames at 480×832) and InfiniteTalk’s 81-frame chunks with nine context frames extend duration without targeting service latency [169, 198].

One Fast implication deserves explicit statement: the conditioning pipeline sets a latency floor that no visual-path acceleration can cross. Speech encoders with wide receptive fields, block-level audio buffering (Teller collects a 200 ms speech block before computing [226]), and window-based speech pre-encoding all contribute delay ahead of the first DiT call; StreamAvatar’s 1.20 s first output includes 0.48 s of pure input buffering [161]. Reports that time only the denoiser therefore understate TTFF by construction, and reproducible streaming protocols should jointly report input-chunk duration, TTFF, steady-state inter-chunk latency and jitter, playback buffer, device count, and module-specific timing endpoints.

4.6 Stable: Evidence for Streaming State Management

The patterns in Section 2.7 act on different failure paths. We now examine their implementations and observed effects in streaming diffusion. Generic long-video work clarifies how these patterns interact. Self-Forcing++ samples five-second segments from long self-generated rollouts for distributional supervision (L4-G at scale), exposing training to states far from the sequence origin; its standardized quantitative horizon is 100 s [40]. LongLive combines short-window attention, a persistent frame sink (L1), KV recaching, and streaming long tuning; its recache is event-triggered at a prompt-switch boundary, rebuilding KV from recent generated video and the new prompt to remove stale semantics [197]. The reported 20.7 FPS uses a 1.3B model at 832×480 on one H100, with quantitative tests at 30 s (single prompt) and 60 s (scheduled prompt switches) and a 240 s demonstration. Mapped to virtual-human state: a short window preserves immediate mouth and motion continuity, an identity sink stabilizes appearance, compressed long-term KV recovers information lost by local attention, event-triggered recaching removes stale semantics, and RoPE re-indexing controls positional extrapolation. Together, these mechanisms show that computation can continue with finite memory; they do not by themselves demonstrate recovery of identity, synchronization, and motion over a long session [161, 197, 227].

From a state-estimation perspective, the patterns have complementary roles: the anchor (L1) supplies identity and scene priors, the recent window preserves velocity and mouth continuity, and long-term memory (L2) retrieves poses or events forgotten locally. An ever-growing cache increases computation and error contamination, and a constant cache can forget remote events. A confidence-based design that updates, compresses, or resets memory adaptively remains an open problem. Stable evaluation should measure identity, appearance, synchronization, and motion in predeclared duration bins, and report error trends, legitimate changes blocked by the anchor, generation faults, and recovery length.

Live Avatar illustrates the joint design of sampling, deployment, and bounded state. It combines an Adaptive Attention Sink (L1), Rolling RoPE (L3), and History Corrupt, which uses step-matched noisy KV and removes a clean-cache call. Its window retains four recent blocks of three latents each and one persistent sink; after the first chunk, the first generated latent replaces the original reference sink. The long-horizon test loops roughly seven-minute speech recordings and measures windows at 0–10, 100–110, 1000–1010, and 10000–10010 s. The reported Q-Align aesthetic score, image quality, Sync-C, and DINOv2 cosine similarity

Table 6 Compact temporal evidence for diffusion systems. The operation column records architectural presence (A). Q denotes qualitative examples; T denotes quantitative evidence over the stated duration, which may be aggregate or sampled rather than continuously binned. These system-level results do not attribute benefit to every listed mechanism. Component ablations are identified separately in the checked ledger.

Method	State / operation (A)	Output dence	evi-	Reported horizon and limit
Hallo2 [39]	Patch-drop/noisy history (L4-C)	T		4 min, 512 ² , enhancer off; hour-long claim separate
StableAvatar [169]	Reference anchor; overlapping windows	T, aggregate		Long100: 100 clips, 2–5 min; offline generation
PersonaLive [102]	Selected history; capacity/eviction NR	T, Q		LV100: at least 1 min; 2700-frame demonstration separate
MotionStream [153]	Sink (L1), rolling KV, L3, L4-G	T		241 frames (~15 s); general-motion task
JoyAvatar [96]	Boundary transfer (L5), L3	Q		120 s; no time-binned quantitative test
StreamAvatar [161]	Sink (L1), L3, L4-G	T, Q		20-s protocol; ~150-s demonstration
LiveTalk [28]	Three sinks, two recent slots, L4-G	Q		100-s demonstration; several-minute claim
EchoTorrent [124]	Reference sink; tail-only forcing	Q		1000-s demonstration
Live Avatar [76]	Sink (L1), L3, L4-G; noisy KV	T, sampled		Windows up to 10000–10010 s; repeated audio, static-scene prior
LPM 1.0 [211]	Sink/reference; pre-RoPE KV	Q		60,000-s demonstration timeline; time-binned metrics NR
AsymTalker [114]	Asymmetric GT/generated history	T, binned		FID/ArcFace at 5/50/150/300/600 s
Lip Forcing [30]	Sink, dynamic positions, L4-G	T		Up to 6 min; speech pre-encoded
SoulX-FlashTalk [152]	Retrospective self-correction (L4-G)	T, Q		Over 5-min tests; 1000-s demonstration
SoulX-FlashHead [210]	Reference anchor; L4-G	Q		60-s demonstration
SoulX-LiveAct [227]	Fixed reference/long/recent slots (L2)	NR		Hour-scale claim; duration-binned metrics NR
Avatar V [103]	Appearance anchor; motion transfer (L5)	NR		Unbounded-length claim; quantitative duration NR
FlowAct-R1 [179]	Structured memory; generated-history replacement	NR		Longest quantitative rollout NR; full-body scope

do not show marked degradation in those windows [76]. This supports stability at sampled points under repeated audio. The static-scene prior, unsampled intervals, and absence of an external baseline under the same protocol limit broader conclusions.

Interactive systems provide further examples of state management, although interaction capability alone does not establish stability. LPM 1.0 supports speaking, listening, idling, and reflecting and combines a 2-NFE generator with a 1-NFE refiner over pre-RoPE KV (L1+L3). Its benchmark accepts inputs up to one hour and its demonstration timeline reaches 60,000 s, but duration-binned quantitative metrics are NR [211]. FlowAct-R1 uses fixed structured memory and generated-history replacement during training (L2+L4-G) for full-body behavior with MLLM action planning [179]. As a contextual example from the render route, ARIG maintains 512 interaction summaries and reports improved source metrics relative to a capacity of four; this cache capacity is not translated into a measured generation duration [60].

4.7 Design Guidance

Table 7 summarizes three practical considerations. First, the benefit of further step reduction depends on how

much runtime remains in the denoiser; once decoding or conditioning dominates, optimizing those stages may be more effective. There is no universal NFE threshold for this transition. Second, pipeline overlap primarily improves steady-state throughput, whereas smaller initial chunks and shorter input buffers can reduce TTFF. Third, rolling caches, chunk overlap, and memory compression affect both per-chunk cost and the history available for maintaining consistency. Their efficiency and stability effects should therefore be evaluated together.

4.8 Synthesis and Open Questions

Online diffusion-based virtual humans are produced by a stack of mechanisms rather than a universally faster sampler, and a gain at one layer can be cancelled by the next bottleneck. Future work should report the quality–speed–horizon Pareto frontier rather than one favorable scalar, with the model-call ledger (sampler, steps, CFG branches, strict NFE), component timings, TTFF, steady-state chunk time, jitter, and peak memory as distinct fields; cache methods should disclose warm-up, refresh policy, and memory growth with duration.

For stability, the reviewed evidence lacks a matched-compute comparison of all five patterns and their com-

Table 7 Diffusion-route guidance. D marks mechanisms documented in Tables 5 and 6; H marks a proposed transfer or policy. Evidence remains specific to the source configuration.

Target	Design options	Trade-offs to check
Faster fixed clips	D: step caches, CFG folding, compressed latents	Cache staleness near detail steps; compression harms mouths and teeth
Playback-rate streaming	D: few-step DMD, rolling KV, DiT/VAE overlap, FP8 and fused kernels	Multi-GPU budgets; Tiny-VAE quality loss; L4-G training cost
Low TTFF	D: smaller initial chunks and staged startup	Input buffering floor; startup-quality gap
Minute-scale stability	D: anchors, generated-history training, position management	Static-scene bias; motion-diversity loss; memory policy requires validation
Hour-scale operation	D: bounded memory, anchors, positions; H: adaptive refresh/recovery	Evidence depends on sampled windows and input protocol; no duration guarantee

binations. Such a comparison should examine both drift reduction and restrictions on motion diversity or legitimate scene changes. Memory updates based on estimated reliability are a further research direction. Evaluation should track identity, synchronization, visual quality, background drift, motion, and failures in predeclared time bins; training studies should compare ground-truth, generated, and mixed histories under matched conditions. Section 9 also considers authorization and provenance for long-running systems with cached identity state.

Diffusion-based virtual humans combine representation compression, fewer sampling evaluations, computation reuse, and deployment optimization. Streaming methods additionally manage generated history through anchors, memory, position handling, training, and motion transfer. These mechanisms offer complementary benefits, but their joint effect depends on the task and measurement protocol. When a native autoregressive backbone instead predicts the evolving visual state and diffusion supplies only a local emission head, the system belongs to the autoregressive route discussed next.

5 Autoregressive Methods

The term *autoregressive* (AR) appears at several levels of a virtual-human system. A dialogue model may predict speech, a controller may predict motion, and a visual model may predict frame tokens. This section reserves the AR route for the last case. A native AR model must own cross-frame dense or discrete visual state, as in MIDAS and Body of Her [4, 26]. An AR speech, language, pose, landmark, or motion branch does not establish visual ownership when another component owns appearance, and a sequential scheduler does not make a dense diffusion process autoregressive. This narrow definition separates system organization from the mechanism that produces visible temporal content.

The literature nevertheless advances along two largely independent axes. The first is *state ownership*: causal state progresses from language and behavior, through motion and sparse control, to decodable visual representations. The second is *interaction scope*: one-way animation expands toward online dyadic response, full-duplex interaction, and persistent world grounding. A world-model label belongs to the second axis unless the causal model also predicts the evolving visual representation. This distinction permits highly interactive systems to inform AR system design without relabeling their flow-, diffusion-, or renderer-owned visual generation paths.

Autoregressive visual state connects efficiency directly to history management. Reusing context can avoid repeated computation, while a long context increases memory and may retain generation errors. KV caching, quantization, parallel decoding, and speculative execution provide candidate acceleration mechanisms [71, 93]. Their benefits for a virtual-human pipeline depend on the visual representation, local decoder, and quality constraints, which must be evaluated alongside temporal drift.

5.1 Scope, State Ownership, and Cost Boundary

The ordered test in Section 2.3 first excludes methods whose appearance is owned by a subject-specific mesh, neural field, Gaussian set, or identity-conditioned warp. Among the remaining systems, the AR route is selected only when the evolving visual tokens or latents are predicted natively from prior visual state; a downstream diffusion process may refine the current representation without changing that assignment.

For one emitted frame or block, a useful route-specific accounting is

$$C_t = C_{\text{condition}} + C_{\text{AR}}(h_t) + C_{\text{local-emission}} + C_{\text{decode/render}} + C_{\text{system}}, \quad (5)$$

where h_t names the persistent visual state available at step t . The AR term includes token prediction, attention, and state update; a local continuous-output head retains its own denoising or flow evaluations; and the final terms include codec decoding, rendering, transfer, compositing, and input/output (I/O). Consequently, one AR-backbone update per frame does not imply only one network evaluation or one complete system call per visible frame, and a reported frame rate is interpretable only with the hardware, resolution, precision, and included stages defined in Section 2.6.

5.2 Mechanism Map: Compression, State, and Emission

AR visual generation exposes three coupled design choices [4, 26, 191]. First, the representation fixes serial granularity. A frame may contain many discrete tokens [171], a compact set of continuous tokens [23], or one latent block. Spatial compression and within-frame parallel prediction reduce different work, but tokenizer and decoder costs remain. Second, persistent state controls repeated computation and resource growth. KV reuse avoids recomputing past projections; windows, summaries, eviction, and resets can bound the retained state. A codec feature cache is not interchangeable with an AR-backbone context. Third, local emission converts predicted state into an observable frame. A diffusion or flow head may improve continuous output, but its calls and final decoder remain on the critical path [4, 26].

These choices create a characteristic efficiency tension. More aggressive visual compression shortens the AR sequence but may move cost or reconstruction error into the codec. A longer cache supplies more context while increasing memory and preserving erroneous history; truncation bounds memory but can discard identity, pose, or conversational events. Parallel prediction inside one frame reduces serial depth, yet cross-frame self-conditioning still exposes the model to its own errors. Fast and Stable therefore require separate measurements rather than an architectural label.

5.3 Autoregression at Different Pipeline Levels

The predicted variable determines the role of an AR module. UniTalker and OmniResponse model responses and compact facial behavior; Teller, ARIG, DyStream,

Talker-T2AV, and UniLS predict motion or audiovisual controls that a renderer maps to visible appearance [21, 34, 60, 70, 115, 203, 226]. These contextual systems motivate causal control but do not establish a visual AR speed or stability result. Their renderer remains in the system ledger.

Native visual AR instead predicts dense visual latents or appearance-bearing tokens. UniMo interleaves 2D video and quantized 3D motion tokens, providing a general human-video bridge; MIDAS, Body of Her, and EARTalking target conditioned human video with framewise visual state [4, 26, 129, 187]. The inference-graph test and worked cases in Table 2 replace a second taxonomy here.

5.4 Representative Visual-State Autoregression

UniMo, MIDAS, Body of Her, and EARTalking satisfy the visual-state criterion under different task and decoding assumptions. UniMo supplies a general human video-motion example, whereas the other systems target digital humans or talking heads. Their inclusion establishes architectural membership; the completeness of their deployment evidence is assessed separately in Table 8.

MIDAS is the clearest retained example of an AR visual owner followed by a local iterative head. A Qwen2.5-3B backbone predicts the next continuous frame representation, using a Deep Compression Autoencoder (DC-AE) [23] with $64\times$ spatial compression and at most 60 visual tokens per frame; an approximately 0.5-billion-parameter PixArt- α head then applies four denoising steps to the current representation before DC-AE decoding [26]. The six-frame logical block exposes all of its condition tokens to subsequent frame tokens. The implementation separately retains a five-frame DC-AE feature/KV history and truncates AR frame context to the most recent eight blocks. These are two state objects with different owners, and the paper does not disclose the AR cache implementation or reset behavior. On timing, the reported 480 ms block length is media span and the 25 FPS identity-specific output is a media rate; inference hardware, measured FPS, time to first frame (TTFF), and a component ledger are not reported. The paper states that identity-specific post-training can support up to one hour and shows a qualitative four-minute singing case, but neither is accompanied by time-binned stability measurements. MIDAS therefore supports the mechanism claim that compressed AR visual state can drive a four-step local emission head. It does not establish measured end-to-end Fast or Stable performance [26].

Table 8 Native visual AR evidence. Local sampling remains part of the timing boundary. NR refers to the checked source, not to all public material. See Table 2 for the architectural decisions.

Method	State / local emission	Timing and resource evidence	Temporal evidence
MIDAS [26]	Eight AR blocks; five codec frames; four-step local head	Measured FPS, TTFF, hardware NR; 25 FPS/480 ms are media rate/span	One-hour author claim; four-minute qualitative demonstration; time-binned metrics NR
Body of Her [4]	Frame-causal state; compact parallel diffusion head; cache bound NR	42 ms/next frame at 640×360 ; device and included stages NR; 24 FPS media rate	Long-session quantitative duration NR
UniMo [129]	Visual and motion tokens; deployment cache contract NR	General human video-motion bridge; conversational service timing NA	Conversational long-session evaluation NA; general-task results separate
EARTalking [187]	Ten-latent-frame window; 36 spatial AR and 25 denoising steps	512 ² setup; numeric FPS, TTFF and inference device NR; source notes real-time limitation	L1/L4-C ablations (B); Q at $3 \times$ training window; minute-scale binned metrics NR

Body of Her provides a second, framewise instance. Its Transformer decoder conditions on prior human and agent audio-video histories, current text or trajectory controls, and identity, thereby owning cross-frame behavior and visual evolution; a compact diffusion head samples all continuous tokens of the current frame in parallel, after which an unspecified video decoder produces pixels [4]. The paper describes a roughly 8-billion-parameter model emitting 640×360 synchronized audio-video at a 24 FPS media rate and states that the next frame is generated within 42 ms, but reports neither inference hardware nor the diffusion step count, strict NFE, or included components, so the 42 ms statement cannot enter a common speed comparison. Context capacity, cache growth or eviction, reset behavior, and any quantitative Stable duration are also not reported, and the demonstrated customized agents use 9 and 5.5 hours of subject-specific data. The method supports a route and feasibility claim, not a quantified speed, generalization, or long-session claim [4].

EARTalking provides a stronger framewise case. Its GPT-style causal model carries prior visual context and generates the next 3D-VAE frame representation with frame-level control; a sink-frame anchor, sliding KV window, and cyclic position policy make its persistence contract more explicit than those of the earlier retained systems [187]. The frame-local generation and final decoder still belong in the cost ledger, but the evolving decodable visual state is owned by the causal backbone. EARTalking consequently passes the route test without implying direct pixel-space next-token prediction.

The route boundary remains consequential for evidence interpretation. DyStream reports 29.24 FPS and 34.2 ms audio-packet-to-motion delay on one H200, but these are motion-level measurements for a render-owned system [21]. Conversely, native visual owners may disclose a causal state contract without publishing a complete end-to-end timing ledger. Table 8 therefore keeps

route membership separate from the completeness of Fast and Stable evidence.

Full-duplex systems as contextual interfaces. X-Streamer and the Wan-Streamer series separate interaction state from dense visual generation. X-Streamer’s Thinker conditions a video Actor whose backbone iteratively denoises dense chunks [190]. Wan-Streamer v0.2 separates a token-causal Thinker from a context-parallel flow-matching Performer; v0.3 makes relatively persistent world context and time-localized events explicit [73, 74]. These diffusion-route examples show how serving and interaction can be modular without transferring visual ownership to the dialogue model. Successive Wan releases are linked versions, not independent evidence for an AR visual mechanism.

5.5 Fast Design Guidance

The Fast levers below are tied to the cost terms in Equation (5). Evidence from native visual owners supports route-specific claims; motion-, diffusion-, and flow-owned systems are used only to expose serving opportunities or missing component boundaries.

- **Compress the serial visual state** ($C_{AR} \downarrow$). MIDAS limits a frame to at most 60 DC-AE tokens, while UniMo interleaves compressed video and quantized motion tokens. Compression can reduce serial work but may move cost or error into the codec, particularly for fine facial detail and rapid articulation.
- **Parallelize local emission** (serial depth $\downarrow$). Body of Her samples all continuous tokens of the current frame in parallel through a compact diffusion head, and MIDAS uses a four-step local head after the AR frame state. EARTalking similarly separates cross-frame causal state from frame-local realization. The local head’s strict number of function evaluations (NFE), tensor shape, and decoder cost remain part of end-to-end latency.

- **Reuse and bound persistent state** ($C_{AR}(h_t)$ growth control). Standard KV reuse avoids recomputing past projections. MIDAS truncates AR context to eight blocks while separately retaining five codec frames; EARTalking combines a sink-frame anchor with a sliding KV window and cyclic positions. These mechanisms bound repeated work, but their eviction, correction, and reset costs require direct measurement.
- **Separate control-path and emission-path serving** ($C_{system} \downarrow$). Wan-Streamer v0.2 keeps its token-causal Thinker on one GPU and places high-resolution flow-matching generation in a Ulysses context-parallel Performer [73]. This is not native visual AR evidence, but it places the causal state update and the expensive local emitter on distinct serving paths organized by a pipelined unit schedule, without changing the route assignment.
- **Test speculative visual decoding as a transfer hypothesis**. Speculative decoding is established for language models and has been studied for general AR video [71, 93]. A virtual-human deployment must additionally verify identity, lip synchronization, and visual-state acceptance when drafted tokens or frames are committed; none of the retained native visual systems reports that ledger.

The route boundary also exposes bottleneck migration. DyStream reports motion-level packet delay while its warp, decoder, and display boundary remain undisclosed; X-Streamer reports a multimodal system rate without a matched decomposition of its dense diffusion Actor; and Wan-Streamer v0.2 reports both model-side and network-inclusive interaction latency for a flow-owned visual path [21, 73, 190]. These values answer different questions. A reproducible AR report should provide AR calls per visible frame, token or block count, local-head NFE, cache-update cost, decoder time, I/O boundary, and peak state memory under one declared pipeline.

5.6 Stable Design Guidance

Visual autoregression repeatedly consumes committed visual state, which can preserve continuity and propagate error. The common patterns therefore require method-specific checks. EARTalking documents an L1 reference sink, L3 cyclic positions, and L4-C noise-corrupted histories. MIDAS also uses L4-C; its five-frame codec history and eight-block AR context are different state objects and do not alone establish multiscale L2 memory. Body of Her’s dataset-history training is dis-

Table 9 AR-route guidance. D marks documented mechanisms in Table 8; H marks an unvalidated transfer or deployment policy. A mechanism label does not establish the target duration.

Target	Design options	Trade-offs to check
Low serial latency	D: compression and parallel within-frame emission	Codec error on facial detail
High throughput	D: KV reuse; H: speculative visual decoding	Draft-model identity fidelity unknown
Minute-scale stability	D: anchors and bounded history; H: generated-history training and adaptive refresh	Accumulated error; anchor over-constraint; refresh discontinuity; no matched quantitative evidence
Hour-scale operation	H: hierarchical state and declared eviction/reset/recovery	Long-range forgetting; reset discontinuity; unmeasured memory and error trajectories

tinct from generated-history training. None of these facts should be relabeled as an L4-G rollout study.

Explicit boundary transfer (L5) likewise requires a designated motion or boundary variable; ordinary access to preceding visual tokens is insufficient. Generated-history training, adaptive reference refresh, and hierarchical state are plausible research directions for the route (H), not demonstrated solutions shared by all its methods. EARTalking’s source ablations are useful partial evidence, analyzed in Section 7; the retained systems do not yet provide a complete joint time-binned session protocol.

5.7 Synthesis and Open Questions

The reviewed visual AR systems span video-motion tokens, compressed frame representations, and framewise controlled generation. Their tasks, personalization requirements, tokenizers, and local emission heads differ. No system in Table 8 provides both complete end-to-end timing and a joint time-binned stability evaluation. Consequently, architectural feasibility is supported more broadly than a common deployment comparison.

The most useful next step is interface-level co-design: visual tokenizers should report how compression changes both AR work and decoder cost; state managers should expose cache growth, eviction, correction, and reset; local heads should disclose strict NFE; and serving evaluations should charge condition encoding, decoding, and I/O to the same event timeline. Sub-linear-state sequence models [55, 200] are a natural research direc-

tion for the route’s Stable problem, offering constant-size state with different forgetting behavior than KV eviction. Response-, motion-, and world-level AR can be excellent deployment choices, but their appearance owner and emission costs remain assigned to the diffusion, flow, or rendering routes that produce visible content.

Autoregression can organize behavior, motion, or the visible sequence itself. Only the last role defines the visual AR route. Its main design choices concern representation compression, history reuse, and local frame emission; each affects both runtime and error propagation. Future comparisons should evaluate these choices through explicit state-update policies and matched efficiency and stability measurements.

6 Render-Based Methods

6.1 Scope, Ownership, and Cost Boundary

Render-based methods separate a reusable appearance representation from the time-varying controls that animate it. We assign a system to this route when a subject-specific mesh, neural field, Gaussian set, or identity-conditioned image-space representation owns appearance and its renderer produces the visible frame [58, 59, 83]. The controller may itself be deterministic, autoregressive, diffusion-based, or flow-based; generating compact motion with one of these models does not change which component owns image synthesis. This ownership test also distinguishes the present route from direct pixel or video-latent generation in Sections 3–5.

Reusing appearance connects the route’s efficiency and stability mechanisms. An enrolled asset reduces repeated appearance construction and supplies a persistent identity reference, corresponding to L1 in Section 2.7. Frame production still requires deformation, rendering, and often neural decoding or inpainting. Enrollment, tracking, and compositing must therefore be included when assessing the complete pipeline. For a session of T observable frames, we write

$$C_{\text{session}} = C_{\text{enroll}} + \sum_{t=1}^T [C_{\text{track/cond}}(t) + C_{\text{motion}}(t) + C_{\text{repr}}(t) + C_{\text{render}}(t) + C_{\text{compose}}(t) + C_{\text{system}}(t)]. \quad (6)$$

Here, C_{enroll} covers capture, calibration, tracking, reconstruction, rigging, per-subject optimization, and asset storage; $C_{\text{track/cond}}$ covers runtime acquisition and estimation of pose, camera, gaze, audio, or other controls; and C_{motion} is the cost of predicting the compact time-varying state. The remaining terms cover

appearance deformation or attribute updates, rendering, masks/inpainting/super-resolution, and system overhead such as transfer, scheduling, encoding, and I/O. Thus, renderer-only throughput cannot be reported as end-to-end latency, and a high per-frame rate cannot hide a costly enrollment stage.

Figure 7 projects Equation (6) onto four recurring interface choices. It is an ownership and cost audit rather than a measured universal pipeline: enrollment and reused appearance are session-level, whereas the control-to-output path repeats per frame, and individual methods may omit or relocate the optional source-pixel and compositor paths.

Codec avatars provide an early example of person-specific learned appearance for real-time telepresence under studio capture conditions, including deep appearance models [110], expressive extensions [31], and audio- and gaze-driven animation [145]. Subsequent work explores different enrollment budgets, from multi-view capture to monocular video and single-image reconstruction. These settings offer different trade-offs between acquisition effort, subject fidelity, and rendering cost.

6.2 Explicit Controllable Appearance Representations

Explicit backends expose a geometric or scene representation that is updated by pose, expression, viewpoint, or audio-derived controls. Parametric models such as 3DMM, FLAME, SMPL, and the expressive SMPL-X provide interpretable control interfaces [8, 99, 112, 133]; they do not, on their own, establish photorealism or real-time operation. RC-SMPL sits at the lightweight end of this parametric interface: it animates an SMPL-X body mesh whose texture and normal maps are accumulated in real time from a single RGBD camera, trading multi-camera capture for progressive per-subject enrollment; the source claims real-time operation but reports neither frame rate nor inference hardware [157]. Likewise, NeRF and 3D Gaussian Splatting (3DGS) define rendering families [83, 126], while capture coverage, representation size, deformation, compositing, and hardware determine the operating point of a particular avatar.

6.2.1 Neural-field backends

Early talking-head neural fields made the appearance asset target-specific. AD-NeRF learns separate audio-conditioned head and torso fields from a 3–5-minute target video sampled at 25 FPS, under a static-camera/background setting [59]. In its reported five-minute setting, training takes about 36 hours and rendering one 450×450 frame takes about 12 seconds on an RTX 3090. This is a frame-rendering measurement, not

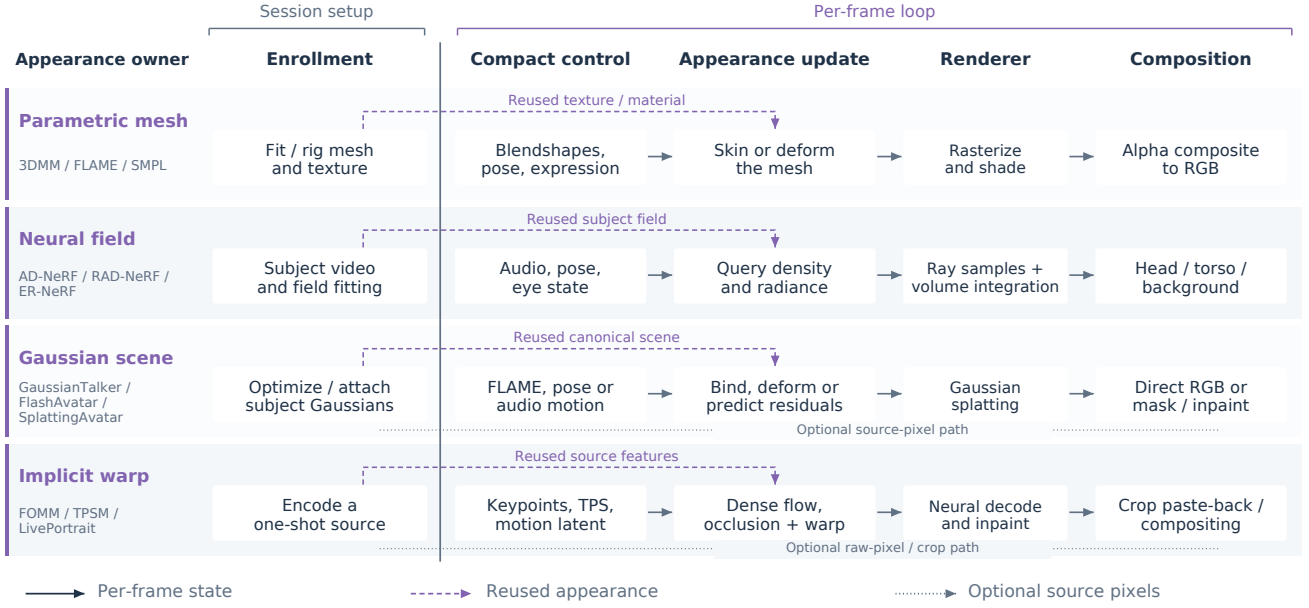


Fig. 7 Interface decomposition of representative render-based appearance owners. Columns separate enrollment from per-frame control, appearance update, rendering, and composition. Solid arrows denote per-frame state; dashed arrows denote reused appearance; dotted arrows denote optional source-pixel paths. The rows summarize parametric, neural-field, Gaussian, and implicit-warp alternatives supported by representative primary papers [58, 59, 98, 99, 155, 188, 207, 221]. They are not a speed, quality, or Stable ranking. 3DMM, FLAME, NeRF, RGB, SMPL, and TPS are defined in the surrounding text.

end-to-end throughput. HeadNeRF instead renders a 32×32 neural feature map and uses a 2D neural renderer to produce a 256×256 image [66]; it reports about 25 ms per frame (over 40 FPS), but the inference GPU is unspecified and the timing begins from supplied latent codes and camera parameters, excluding fitting, tracking, and I/O. FLAME-in-NeRF attaches parametric expression control to the field [5].

Hash-grid and occupancy structures then removed most field-query work without removing subject enrollment. This is the route’s adoption of the Instant-NGP lever, in which a multiresolution hash encoding replaces deep MLP queries [127]. RAD-NeRF combines a 3D spatial hash, an audio feature grid, occupancy pruning, and a separate 2D torso stage; it reports 40 FPS and about 7 hours of target training on one V100, with output resolution and timer coverage not reported in the source runtime statement [166]. ER-NeRF uses tri-plane hash encodings, regional attention, and a separate 2D torso model [98]; its 34 FPS result is reported for a 450×450 Obama comparison on one RTX 3080 Ti, with included runtime modules unspecified. These reports illustrate efficient personalized field designs, but their different devices, timing boundaries, and enrollment procedures do not isolate a common speedup over AD-NeRF.

SyncTalk retains the personalized neural field but makes audio–visual synchronization the central design target. A tri-plane hash head field is driven by a Face-

Sync controller aligning lip motion to audio, a 3D facial-blendshape expression model, and a Head-Sync stabilizer for pose, and a Portrait-Sync generator restores hair detail and blends the head with the torso [136]. It reports 52 FPS without and 50 FPS with the Portrait-Sync stage at 450×450 on one RTX 3090 with preloaded data, after about two hours of per-subject training. The synchronization machinery is a quality lever on the conditioning path rather than a change to C_{render} ; its Gaussian successor SyncTalk++ appears in Section 6.2 below.

Beyond talking heads, AvatarReX extends the personalized neural field to expressive full-body avatars with separately controllable body, hands, and face, learned from multi-view video [228]. Its efficiency lever is a deferred rendering pipeline built on surface rather than dense volume rendering; the source reports 25 FPS at 1024×1024 on one RTX 3090 for animation and rendering, characterized as roughly two orders of magnitude faster than prior NeRF avatars. Enrollment is studio-scale (22 synchronized cameras, about 2000 frames per sequence), the opposite end of the C_{enroll} spectrum from the one-shot methods below.

6.2.2 Gaussian scene representations

Gaussian avatars replace repeated field sampling with splattable primitives, but they differ substantially in ac-

quisition and animation. GaussianTalker binds a target-specific Gaussian face to FLAME and predicts speaker-specific blendshapes; Gaussian rasterization and an inpainting generator synthesize the face, while pixels outside the facial mask are retained from the source video [207]. Its two training stages take roughly 6 and 4 hours on one RTX 4090. The paper reports 130 FPS on that GPU with the sequence preloaded and 36 FPS for an MNN/OpenGL implementation on an Apple M1, an early sign that splatting fits mobile compute budgets. GaussianAvatars optimizes a subject from multi-view observations and assigns Gaussians to FLAME triangles so that they inherit mesh motion [140]. SplattingAvatar uses a registered mesh for low-frequency deformation and walkable barycentric Gaussians for higher-frequency appearance; its Unity implementation reports over 300 FPS on an RTX 3090 and a steady 30 FPS on an iPhone 13 (resolution and timer window unreported) [150]. FlashAvatar narrows the personalized head to 13,453 surface-embedded Gaussians deformed by tracked FLAME motion and a residual MLP, reporting 304 FPS at 512×512 on one RTX 3090 for animation and rendering only [188]. A broader family varies the binding and deformation design without changing the cost structure: HeadGaS’s blendshape-latent formulation [47], PSAvatar’s point-based shape model [225], deformable body avatars [141], emotional control [196], progressive construction [158], and real-time full-pipeline avatar systems [168], text-driven toonification combining a conditional tri-plane with a Gaussian deformation field (TextToon: 48 FPS on one RTX 4090 and 15–18 FPS on an Apple M1) [159], and real-time hand avatars fit from as few as 5–20 views (GaussianHand: up to 125 FPS, inference hardware NR) [222]. In every case, sparse, rasterizable appearance shrinks C_{render} , while enrollment, tracking, and topology handling remain.

Audio-driven per-subject splatting has become its own competitive subfamily. GaussianTalker (Cho et al.) encodes canonical Gaussian attributes into a shared spatial-aware feature and predicts per-frame Gaussian offsets through a spatial-audio attention module driven directly by speech, reporting 98 FPS (and up to 120 FPS for a lightweight variant) at 512×512 after about one to 1.5 hours of per-subject training, with the inference GPU NR [29]. This is a distinct method from the FLAME-bound GaussianTalker of Yu et al. [207] despite the shared name: the former drives raw Gaussians from audio, whereas the latter animates FLAME-bound blendshapes. SyncTalk++ ports the synchronization-centric SyncTalk design onto a 3D-Gaussian dynamic portrait renderer and adds an out-of-distribution audio expression generator and a torso restorer, reporting 101 FPS at 512×512 on one RTX 4090 after about 1.5 hours of

per-subject training [135]. The pair illustrates a transition from neural fields to Gaussian rendering within a related synchronization-oriented line of work. The changed hardware, resolution, and auxiliary modules prevent attributing the difference in reported frame rate solely to the renderer.

Recent audio-driven Gaussian work makes the controller–appearance boundary more expressive. GaussianSpeech couples an audio-conditioned Transformer to personalized 3D Gaussians, including expression-dependent attributes and multi-view supervision; DGTalker instead navigates a disentangled Gaussian latent space that separates lip motion from broader talking expression. Both retain a Gaussian appearance owner, so their learned audio or latent modules are controllers rather than diffusion-route owners. GGTalker addresses the enrollment–generalization trade with a generalizable Gaussian prior followed by identity-specific adaptation, making explicit the cost of replacing per-subject optimization with a reusable prior. AudioAvatar extends the same direct audio-to-deformation idea to a personalized whole-body avatar: audio-conditioned particle/deformation fields drive facial, hand, and body dynamics before differentiable Gaussian rendering. Its feature-level distillation from a video diffusion model is a training or controller mechanism; the Gaussian representation remains the primary appearance owner. These systems therefore move beyond the conventional audio–pose–renderer interface, but they also move stability risk into OOD audio, large motion, multi-view consistency, and deformation quality [3, 69, 92, 104].

GaussianHeadTalk studies a complementary temporal failure mode: inconsistent Gaussian mapping and noisy facial tracking can produce wobble despite a fixed identity asset [1]. This motivates separate measurements of tracking and deformation error, failed detections, and temporal artifacts, alongside identity and synchronization.

Two recent developments extend the Gaussian recipe along the axes this survey tracks. First, *mobile and wearable deployment*: TaoAvatar delivers full-body talking avatars on augmented-reality glasses by baking a StyleUNet-learned deformation into a lighter MLP and reports 90 FPS on the Apple Vision Pro [24], and SqueezeMe distills multiple Gaussian full-body avatars to run simultaneously on a Quest 3 headset by moving corrective computation out of per-frame networks [77]. These examples support operation on wearable devices under the reported conditions; sustained power and thermal behavior require separate measurement. Second, *one-shot Gaussian construction*: LAM builds an animatable Gaussian head from a single image in one feed-forward pass, explicitly targeting cross-platform real-

time animation and rendering [63]; GUAVA extends fast feed-forward construction to upper-body avatars [212]; and VASA-3D lifts VASA-1’s motion space to a Gaussian head built from one portrait [193]. One-shot construction attacks the route’s historical weakness, hours of C_{enroll} , by amortizing enrollment into a generalizable network. The reduced input coverage leaves unseen appearance and subject-specific detail to the learned prior, creating a fidelity trade-off that depends on the task.

The generalizable head-avatar line makes the enrollment-versus-representation trade especially legible through two works by the same authors. GPAvatar reconstructs a controllable head from one or several images in a single forward pass using a tri-plane feature field with NeRF-style volume rendering, driven by a FLAME point-based expression field and a multi-image attention module; it renders a 128×128 feature map super-resolved to 512×512 and reports about 15 FPS on an A100, which the authors explicitly label pre-real-time [33]. GAGAvatar keeps the one-shot, generalizable, FLAME-conditioned formulation but replaces the volumetric field with 3D Gaussians reconstructed by dual-lifting, reporting 67 FPS at 512×512 on an A100 using 2.5 GB of memory (driving-render only, excluding pre-computable driving parameters) [32]. Both use an A100 and report 512×512 output, but they change the reconstruction and rendering pipeline together. Their timing difference is therefore not a controlled ablation of representation alone. GAGAvatar also serves as a reusable backend for controllers such as REA-Listener (Section 6.4).

6.3 Implicit Image-Space Appearance Reuse

Image-space methods store identity primarily in a source image or its encoded features and predict correspondences to each target pose. FOMM estimates ten keypoints and local Jacobians, constructs dense backward flow and an occlusion map, and uses an inpainting decoder for content that cannot be copied directly [155], building on earlier keypoint-based transfer [154] and refined by articulated region modeling [156]. TPS Motion Model replaces the local affine model with ten thin-plate-spline transformations plus an affine background transformation, combined by multi-resolution masks before inpainting [221]. Both were evaluated as source-to-driving-video animation without runtime disclosure. The lineage acquired deployment weight with face-vid2vid, which learns a 3D-aware keypoint representation for one-shot talking-head synthesis explicitly motivated by video-conferencing bandwidth: transmitting keypoints instead of pixels makes the warp both

the codec and the renderer [182]. Latent-space navigation offers an alternative implicit interface [183], and audio-driven variants animate a still image directly from speech [180].

LivePortrait provides an efficiency-oriented instance of this family. It learns implicit keypoints and motion, warps a source feature volume, and decodes with a SPADE-based generator, supporting cross-identity animation and paste-back compositing [58]. The paper reports 12.8 ms for model generation/inference on one RTX 4090 in naive PyTorch, without binding the timer to a stated resolution or decomposing preprocessing, paste-back, and I/O. Its one-shot appearance interface is reused by several systems discussed in Section 6.4. Large pose changes, disocclusion, mouth interiors, hair, and crop-boundary compositing remain relevant failure cases when evaluating these composed systems.

6.4 Motion Controllers over Reused Appearance

Reusable appearance permits motion control to be studied separately from image synthesis. A compact motion model can drive a one-shot warp or Gaussian asset, changing temporal behavior without changing the visual owner. The controller and renderer have separate costs, and either can determine system latency.

VASA-1 illustrates this separation. A diffusion Transformer generates holistic facial dynamics and head motion in a learned face latent space, conditioned on audio and optional control signals, and drives a one-shot appearance decoder; the paper reports online generation of 512×512 video at up to 40 FPS with 170 ms/95 ms latency on one/two RTX 4090 GPUs under its stated protocol [192]. Sampling compact motion can reduce the size of the iterative state, while the appearance decoder reuses identity information. The cost relative to dense-video generation still depends on the networks, sampling schedule, and decoder. FLOAT replaces the sampler with motion-latent flow matching [87]; Ditto structures the motion space for controllable real-time synthesis [100]; ChatAnyone stacks hierarchical motion diffusion for stylized upper-body video with real-time intent [139]; MirrorMe reports real-time audio-driven half-body animation [122]; and FlowTalk pushes motion-space flow matching toward the same operating point [44]. MoDA stays in this compact-controller family while keeping a diffusion motion generator: it reuses LivePortrait’s disentangled facial representation as the appearance renderer and generates motion in a joint parameter space with a rectified-flow model, reducing denoising from 50 to 10 steps and reporting a real-time factor of 0.846 on in-the-wild inputs (below one indicates faster than playback; inference GPU NR) [101]. It is one-shot from a single

reference image, and under the ownership test its LivePortrait renderer, not the motion diffuser, owns appearance. GAIA scales a zero-shot two-stage design (motion generation, then appearance) [62], JoyVASA extends the recipe to animal faces [17], and style-controllable variants [119, 181] enrich the control space. The compact interface also connects to speech-to-3D animation research that predicts vertices or coefficients without owning pixels [51, 223]; such motion-only models are boundary components under the ownership rule.

Interactive-agent systems assemble the same parts with session state. Teller processes 200-ms audio blocks with an autoregressive residual-vector-quantized motion model; an implicit-keypoint warper and a single-pass enhancer produce RGB, with reported stage times of 7, 106, and 71 ms on a machine containing four H800 GPUs [226]. ARIG predicts continuous head-motion coefficients with a 15-step DiffusionMLP over a framewise autoregressive controller and renders through LivePortrait-style warping; it reports a 31 FPS generation-process rate (hardware and included components NR) and maintains a 512-slot interaction summary cache, a bounded controller memory whose Stable effect is unmeasured [60]. DyStream couples an AR/flow motion module (60 ms speaker-audio buffer) to a reference-conditioned appearance volume and reports 29.24 FPS with 34.2 ms audio-packet-to-motion delay on one H200; the delay ends at facial-motion generation rather than final rendering [21]. REA-Listener predicts listener motion with a lightweight Transformer and renders with GAGAvatar [32]/3DGS plus inpainting [224]. Its inherited component-timing extraction is retained in the supplement as unverified because the primary PDF was unavailable during the revision audit; we do not use it to derive a new quantitative finding. PortraitDirector adds hierarchical disentanglement for controllable real-time reenactment [78], and VASA-Rig connects audio-driven rigging to virtual-reality use [128].

6.5 Fast Design Guidance

The route’s levers map onto Equation (6) as follows:

1. **Amortize appearance at enrollment** ($C_{\text{repr}}/C_{\text{render}} \downarrow$ via $C_{\text{enroll}} \uparrow$). Subject-specific fields and Gaussian sets reuse fitted appearance across frames. One-shot warps and feed-forward Gaussian builders, such as LivePortrait and LAM, reduce preparation cost but rely more on learned priors for unseen content.
2. **Structure the queries** ($C_{\text{render}} \downarrow$). Hash grids and occupancy pruning reduce field-query work, while

Gaussian representations permit rasterization. Mobile and wearable examples include SplattingAvatar, TaoAvatar, and SqueezeMe. Their reported frame rates establish particular device configurations, not a measured speed-per-watt ranking.

3. **Keep the controller compact** ($C_{\text{motion}} \downarrow$). Motion-space diffusion or flow reduces the representation processed by an iterative sampler. The renderer and compositor remain on the inference path and require their own component measurements.
4. **Reuse source pixels** (C_{compose} trade). Paste-back and outside-mask retention (GaussianTalker, LivePortrait) avoid re-synthesizing static content, at the price of seams and crop-boundary failures.

Table 10 shows why these levers must be evaluated separately: reported quantities span renderer-only throughput, animation-plus-render time, model-generation time, motion-level packet delay, and block latency, and no cross-paper rank is supported when anatomy, personalization, device, resolution, and compositing differ. The route-specific accounting rule is to include $C_{\text{track/cond}}$ wherever runtime face tracking, FLAME fitting, keypoint estimation, or audio conditioning is required. Precomputed controls and runtime-estimated controls define different boundaries.

6.6 Stable Design Guidance

An enrolled appearance asset provides a persistent identity anchor (L1 in Section 2.7), but does not guarantee stable rendered output. Tracking, controller state, deformation, disocclusion, and compositing can introduce temporal errors even when the canonical asset is unchanged. Controller memories, such as ARIG’s 512-slot cache, provide another form of persistent state. Position handling and training with generated histories are relevant when these controllers are sequential; their longitudinal effects are not established by renderer throughput or asset reuse alone.

The evidence base matches this asymmetry. The duration-structured result highlighted in Table 10 is DyStream’s 180-s head-pose drift ablation, which shows its appearance anchor reducing pose drift over ten three-minute clips [21]. This is a component ablation of anchor conditioning for pose drift, rather than a validation of every L1 mechanism or a full stability protocol. The inherited cohort does not document a joint, duration-binned evaluation of identity, appearance, synchronization, geometry, flicker, memory, and latency. Its incomplete extraction should not be read as proof that every public edition omits each of these measurements.

Table 10 Selected rendering configurations, shortened from the 23 rendering records in the inherited cohort (including the corrected StyleAvatar assignment). The supplement retains full extraction notes. Frame/render time is not TTFF; omitted pipeline stages remain explicit.

Method	Appearance / enrollment	Source timing / device	Boundary or temporal evidence
AD-NeRF [59]	Head/torso fields; 3–5-min capture	~12 s/frame, 450 ² , RTX 3090	Frame rendering; ~36 h training
StyleAvatar [178]	Person-specific UV/background; adversarial decode	35 FPS / 28 ms live; RTX 3090	20 ms model-only; hours-scale enrollment; longitudinal metrics NR
FlashAvatar [188]	FLAME-surface Gaussians; 1–3-min capture	304 FPS, 512 ² , RTX 3090	Animation/render only; tracking excluded
TaoAvatar [24]	Full-body Gaussians; multi-view capture	90 FPS, Apple Vision Pro	Complete pipeline coverage NR
LivePortrait [58]	Source feature volume; one-shot	12.8 ms model generation; RTX 4090	Warping/decoding; preprocessing and paste-back separate
VASA-1 [192]	Reused face appearance; one-shot	Up to 40 FPS, 512 ² ; RTX 4090	170/95 ms for one/two devices under source protocol
Teller [226]	Source image features	7/106/71 ms components; four-H800 machine	Total < 200 ms; TTFF and full RGB boundary NR
DyStream [21]	Reference appearance volume; one-shot	29.24 FPS; 34.2 ms packet delay; H200	Timer ends at motion; 180-s pose-drift ablation
AvatarReX [228]	Body/hand/face fields; 22-camera capture	25 FPS, 1024 ² , RTX 3090	Animation/render; enrollment and tracking separate
GAGAvatar [32]	Feed-forward Gaussian head; one-shot	67 FPS, 512 ² , A100	Driving/render; driving-parameter acquisition excluded
MoDA [101]	Motion flow + LivePortrait; one-shot	RTF 0.846; inference device NR	Wild-data protocol; training hardware is not inference hardware

A useful comparison would evaluate a tracked-and-rendered avatar and an anchored dense generator under the same long-session conditions. Identity retention, motion and deformation error, temporal wobble, and compositing failures should be reported separately. GaussianHeadTalk motivates the distinction between identity similarity and temporal wobble [1]; the reviewed evidence does not establish which complete pipeline is more stable over an hour.

6.7 Synthesis and Open Questions

The central trade-off is not simply fidelity versus throughput. Subject-specific neural fields and Gaussians amortize a substantial enrollment cost over many frames; one-shot feature warpers reduce enrollment but depend heavily on source-view coverage and learned disocclusion; parametric controls provide stable, editable interfaces whose base space cannot express motion it does not encode. Explicit scenes must handle topology, primitive count, memory, and view coverage. Image-space reuse must synthesize newly visible content and preserve crop seams.

Three directions follow from these trade-offs. First, joint design of feed-forward appearance construction and compact motion control could reduce both preparation and runtime costs; one-shot construction still has a nonzero setup cost. Second, controller training should

Table 11 Render-route guidance. D marks a documented mechanism in Table 10; H marks an unevaluated deployment condition. Device-specific results do not establish a route-wide optimum.

Target	Design options	Trade-offs to check
Per-subject reuse	D: Gaussian assets and baked deformation	Hours of enrollment; capture requirements
One-shot generality	D: LivePortrait warps, LAM Gaussians, VASA-1 motion controller	Disocclusion, large poses, mouth interiors; limited input-view coverage
Mobile/wearable deployment	D: splatting and baked correctives (TaoAvatar, SqueezeMe)	Per-subject assets; thermal/battery limits untimed
Minute-scale stability	D: enrolled L1 asset; H: joint long-session reliability	Temporal errors in deformation and composition; joint longitudinal evidence limited
Expressive control	D: motion-space diffusion/flow controller	Controller expressiveness bounded by the asset’s deformation space

be evaluated under generated histories to test whether it reduces drift in recurrent systems such as ARIG and DyStream. Third, matched evaluations should report enrollment data and time, asset size, hardware, precision, all runtime components, and quality over duration. These measurements would show whether a change reduces complete-system cost or shifts work to another stage.

Render-based methods reuse an explicit scene or source-conditioned appearance representation under time-varying control. Efficient queries, compact controllers, and appearance reuse support reported desktop, mobile, and wearable deployments. The reused asset anchors identity, while tracking, deformation, and compositing remain independent sources of temporal error. The comparison in Table 10 therefore records enrollment, runtime, and stability evidence separately.

7 Cross-Route Synthesis from Source Evidence

The route chapters identify mechanisms; this section asks what their evidence permits us to conclude. We use three focused within-source analyses, followed by an architectural synthesis. None pools measurements across unmatched devices or tasks. Exact editions, locations, and configuration records accompany the evidence sheet.

7.1 A Faster Decoder Can Change the Quality Trade-off

PersonaLive isolates a concrete decoder choice in its appendix, Table 4 [102]. Replacing SD-VAE with TinyVAE increases the reported rate from 15.8 to 20.0 FPS, a derived 26.6% increase. Under that same ablation, identity similarity decreases from 0.6983 to 0.6758, average expression distance increases from 0.7028 to 0.7593, and temporal LPIPS increases from 12.83 to 14.66. These measurements establish a local speed–quality trade-off, rather than a universally preferable decoder.

The deduction is that reducing denoiser work changes the value of optimizing other stages. A decoder replacement can improve total throughput while altering the very identity and temporal properties the system is meant to preserve. This experiment does not provide a complete component-time breakdown, so it does not by itself prove that decoding dominates runtime. The useful design rule is to profile the remaining path and measure quality at the chosen operating point.

Table 12 PersonaLive decoder ablation (arXiv v1, supplementary Table 4, PDF p.14). These are source results under one protocol, not cross-paper comparisons. ID is identity similarity; tLP is temporal LPIPS.

Decoder	FPS ↑	ID ↑	tLP ↓
SVD-VAE	11.4	0.6920	12.97
SD-VAE	15.8	0.6983	12.83
TinyVAE	20.0	0.6758	14.66

Table 13 Live Avatar serving configurations (arXiv v6, Table 3, PDF p.12; 720 × 400). TTFF excludes networking. NFE includes the cache-refresh call. †: Table 10 gives 0.94 s for the stated default three-latent block; this configuration discrepancy remains unresolved.

Configuration	GPUs	NFE	FPS	TTFF (s)
Teacher	1	80	0.29	45.50
Distilled	1	5	3.66	4.56
Sequence parallel	4	5	4.50	3.94
Timestep pipeline	4	4	10.16	4.73
+ VAE parallel	5	4	20.88	2.89
+ kernel optimization	5	4	45.20	1.21 [†]

7.2 Throughput and Startup Follow Different System Paths

Live Avatar’s efficiency ablation changes several deployment terms in controlled stages [76]. We use arXiv v6, Table 3, which specifies H100 hardware. Distillation changes the one-GPU rate from 0.29 to 3.66 FPS. With four GPUs, timestep-forcing pipeline parallelism reaches 10.16 FPS but increases TTFF from 4.56 to 4.73 s relative to the one-GPU distilled configuration. Adding a fifth GPU for VAE parallelism reaches 20.88 FPS; the final kernel-optimized configuration reaches 45.2 FPS and 1.21 s TTFF (Table 13).

The measured conclusion is that the final operating point depends on both algorithm and serving topology. The 12.3× ratio between 45.2 and 3.66 FPS is a system-configuration ratio, not a same-resource algorithmic speedup. The pipeline stage illustrates how greater throughput can coexist with a longer startup interval. Even the final kernel row bundles several implementation changes and does not isolate the contribution of each kernel.

Enrollment changes the accounting horizon. For additive resource cost, the amortized cost per frame is $\bar{C}(N) = C_{\text{enroll}}/N + C_{\text{runtime}}$. As an illustrative calculation on one fixed device, a reusable system with 3600 GPU-seconds of enrollment and 0.02 GPU-seconds per frame breaks even with a 0.10-GPU-second direct generator after $3600/(0.10 - 0.02) = 45,000$ frames. This is 30 minutes of content at 25 FPS. The numbers are hypothetical; the calculation shows why enrollment feasi-

bility, expected reuse, and representation coverage must all enter a deployment decision.

7.3 An Ablation Supports a Mechanism Within Its Tested Horizon

EARTalking provides a component-level test of history training and anchoring [187]. Its Table 3 compares corrupted-history training (L4-C) and reference anchoring (L1). The complete model has FID 18.981, FVD 363.909, and Sync-C 5.707; removing corruption gives 20.643, 401.354, and 4.277; removing the anchor gives 42.298, 489.197, and 4.067. The accompanying qualitative example extends to three training-window lengths. This supports the two components under the source protocol, while numeric inference throughput, TTFF, hardware, and minute-scale time-binned stability remain NR in the checked v1 paper.

This case separates three statements that should not be collapsed: a mechanism exists (A), disabling it worsens source metrics (B), and its benefit persists over a particular deployment duration (T). The first two are supported here; a full long-session claim is not. Similarly, Live Avatar’s sampled windows under repeated audio support a specific extrapolation test, not every intervening instant or every changing-input session. Bounded state establishes a resource property, while the outputs establish the tested stability property.

7.4 Reusing Appearance Leaves Other Failure Paths

Rendering separates identity evidence from motion and observation errors. DyStream’s anchor ablation evaluates head-pose drift across ten three-minute clips [21]; GaussianHeadTalk measures temporal wobble and illustrates mouth-detail errors under shorter reenactment protocols [1]. These are valuable partial tests. They address different failure variables and do not supply interchangeable evidence about identity, synchronization, or hour-long service reliability.

The architectural deduction is that a persistent appearance representation narrows what must be regenerated but leaves the controller, tracker, deformation field, and compositor responsible for temporal behavior. The corresponding research hypothesis is that generated-history training and error-triggered refresh could improve sequential controllers without changing the appearance owner. This proposed transfer requires controlled testing. Figure 8 grounds the failure vocabulary in source-reported examples rather than schematic degradation.

8 Datasets and Evaluation

8.1 Dataset Evidence for Generation and Long-Horizon Evaluation

A dataset determines which conditional relationships a model can learn and which claims an evaluation can support. For efficient virtual-human generation, the useful axes are therefore not venue and year, but the generated anatomy, conditioning modalities, identity and interaction structure, media properties, split policy, and continuous observation horizon. These axes expose task mismatch: a high-quality face-video corpus may support appearance learning without providing audio, a multi-view capture may support reconstruction without cross-identity generation, and a dyadic collection may represent turn taking without providing a single continuous session long enough to test drift.

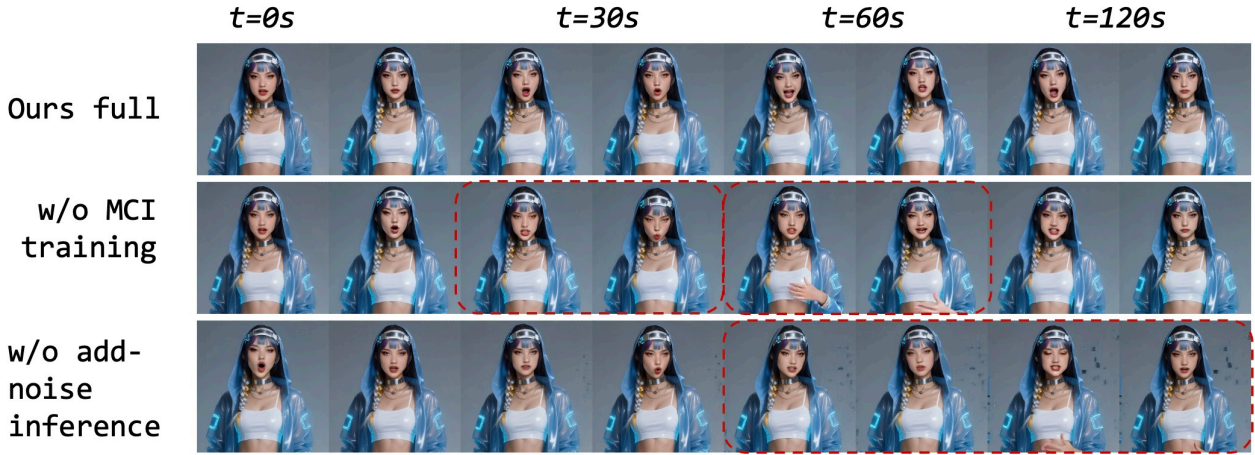
Single-person resources span distinct purposes. HDTF provides in-the-wild high-resolution audiovisual portraits, whereas VFHQ is a video-face super-resolution corpus whose source paper does not establish retained or synchronized audio [189, 220]. CelebV-HQ instead targets unconditional face-video generation and attribute editing, with manually annotated appearance, action, and emotion attributes but no source-defined audio-driving protocol [234]. MEAD adds controlled speech, synchronized views, discrete emotions, and intensity levels [176]. TalkVerse scales synchronized single-person audio–video curation and pose annotation, but its source clips remain short [185]. OpenHumanVid supplies large-scale human-centric video with caption, skeleton, and audio conditions for pretraining human video generators [97], and EchoMimicV2’s half-body corpus extends conditioning to hand and gesture coverage [125]. These resources are complementary rather than interchangeable benchmarks for lip synchronization, identity, expression, or temporal behavior.

Interaction and reconstruction data introduce different supervision. ViCo pairs a visible and audible speaker with a listener’s non-verbal facial response and supplies both seen-identity and identity-disjoint test regimes [232]. SpeakerVid-5M organizes single-speaker clips into dialogue, listening, and multi-turn structures, but its pretraining/supervised fine-tuning (SFT) tiers are quality strata rather than a documented leakage-free train/test partition [216]. Seamless Interaction records long active segments from two participants with separate audio channels and participant-level partitions, while NeRSemble supplies synchronized high-resolution multi-view head capture for sequence-specific novel-view reconstruction [2, 89]. For this survey’s lenses, the relevant property of dyadic data is its *continuous-horizon*

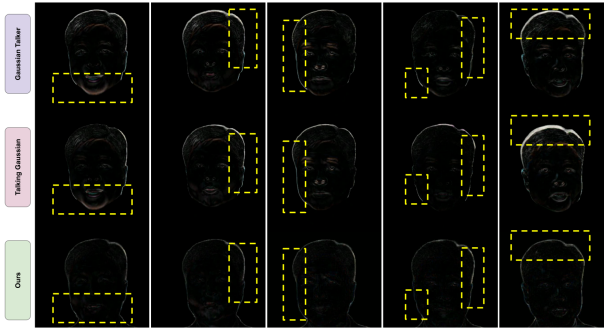
(a) Identity and appearance failure: component ablation



(b) Boundary flicker and later artifacts: explicit motion transfer



(c) Temporal wobble



(d) Mouth and teeth detail



Fig. 8 Observed examples reproduced from primary papers; no examples were generated for this survey. (a) Live Avatar v6, Fig.4, reports identity/appearance degradation when components are removed; the displayed frames have no source timestamp [76]. (b) JoyAvatar, Fig.7, reports block-transition flicker without Motion Condition Injection and later artifacts without conditioning noise; the source labels 0, 30, 60, and 120 s [96]. Static samples illustrate the reported effects but cannot display flicker directly. (c,d) GaussianHeadTalk, Figs.3–4, shows image-difference wobble at ten-frame intervals and mouth-detail comparisons; absolute timestamps are NR [1]. Source labels and comparison methods are preserved. Suppression of an intended change by an anchor remains a separately proposed test, not an observed panel in this gallery.

Table 14 Evidence-bearing properties of representative datasets. Media frame rate is a capture or preprocessing rate, not model throughput. Aggregate corpus duration is not a continuous generation horizon. NR denotes not reported in the primary paper.

Dataset	Task and modalities	Scale and media	Split evidence	Continuous-horizon boundary
HDTF [220]	In-the-wild audiovisual talking face	362 videos; 15.8 h; 300+ subjects; 10k+ sentences; 720p/1080p	Fixed split and identity separation NR	Longest clip NR; 15.8 h is aggregate corpus time
VFHQ [189]	Video-face super-resolution; audio retention NR	16,827 clips from 7,228 source videos; roughly 700–1000 px faces; 512 ² , ×4 benchmark	50 test sequences exclude corresponding training clips; identity separation NR	Tracks retain 100–2000 frames, but frame rate and total hours are NR
CelebV-HQ [234]	Face-video generation and 83-attribute editing; audio NR	35,666 clips; 15,653 identities; at least 512 ² ; source reports approximately 65 h in text versus 68 h in a table	Fixed identity-disjoint split NR	Each clip is 3–20 s and is truncated at 20 s
MEAD [176]	Controlled emotional audiovisual talking faces; seven synchronized views	60 actors; 281,400 clips; 1920×1080 at a 30 FPS capture rate; 48-kHz audio; neutral plus seven emotions and three non-neutral intensities	Paper-specific 20% holdout; split unit and identity separation NR	38 min 57 s per speaker aggregates many short utterances, not one session
ViCo [232]	Dyadic speaker audio-video to listener facial response	483 pairs; 92 identities; 95 min 22 s; 384 ² face crops; source video 1080p at a 30 FPS capture rate and 44.1-kHz audio	D_{test} identities occur in training; D_{ood} is identity-disjoint	Clips span 1–71 s; no minute-scale drift protocol
SpeakerVid-5M [216]	Single, dialogue, listening, and multi-turn audiovisual organizations	Single branch: 5.2M clips/8.7k h; dialogue branch: 770k pairs/1.8k h; branches are not proven disjoint	Pretraining/SFT are quality tiers; VidChatBench has 500 unseen-speaker pairs, but global source/identity leakage control is NR	Basic clips span 3–14 s; stored order does not establish a continuous long-horizon sample
TalkVerse [185]	Single-person synchronized audio-video with 2D poses and structured annotations	2.3M clips; 6.3k h; 720p/1080p; unified to a 25 FPS content rate	Fixed identity-disjoint split NR	Source clips span 5–50 s; aggregate hours do not establish long-session samples
OpenHuman Vid [97]	Human-centric video with caption, skeleton, and audio conditions	13.2M clips reported by the source; large-scale pretraining corpus	Source-defined filtering tiers; identity-disjoint split NR	Clip-scale samples; not a continuous-session resource
NeRSemble [89]	Multi-view human-head capture and sequence-specific novel-view synthesis	222 participants; 4,734 sequences; 31.7M camera images; 7.5 h; 16 views at 3208×2200 and a 73 FPS capture rate	Dataset split NR; experiments use 12 input and four held-out views within each sequence	Roughly 3 min per participant is divided among 25 short sequences; longest sequence NR
Seamless Interaction [2]	Dyadic speech, face, upper-body, hand, transcript, and interaction metadata	4,065.04 h; 64,739 interactions; 5,098 sessions; 4,284 participants; dual 48-kHz audio and 2160×3840 video at a 30 FPS capture rate	Public train/dev/test plus private test; participant partition is functionally clique-level	Released active segments span 2–10 min; one-hour sessions include unreleased meta time; generation evaluation uses about 20 s

Abbreviations: SFT, supervised fine-tuning; D_{test} and D_{ood} , test and out-of-distribution subsets; 2D, two-dimensional; p, progressive-scan resolution; px, pixels; k, thousand; M, million; kHz, kilohertz; h, hours; min, minutes; s, seconds; dev, development; FPS, frames per second.

value: long uninterrupted sessions are exactly the material a stability protocol needs, independent of any interaction claim.

Table 14 distinguishes corpus scale from continuous observation length. Capture or normalized frame rates describe the data rather than generation throughput, and aggregate hours can combine many subjects, views, or short clips. The listed resources do not establish a matched hour-long continuous-generation benchmark. Some long-horizon evaluations therefore extend shorter inputs through repetition, as in Live Avatar [76], which limits the diversity of conditions tested. A public benchmark of continuous 10–60-minute single-subject and dyadic sessions would help measure drift under sustained, changing inputs.

Dataset governance is also part of evaluation validity. Web-derived collections report filtering and source-tracking procedures, but public availability of source videos does not by itself establish participant consent or redistribution rights [185, 189, 216, 232]. Controlled captures provide stronger acquisition records: NeRSemble

reports General Data Protection Regulation (GDPR)-compliant participant agreements and deletion rights, and Seamless Interaction reports adult informed consent and compensation [2, 89]. Their primary papers nevertheless do not specify a complete dataset license. Reproducible reporting should therefore separate media provenance, consent, access terms, and technical split integrity rather than compressing them into a binary *public* field.

8.2 Evaluation Scope and Protocols

Evaluation must preserve the distinction introduced in Section 2.1: visual quality, identity fidelity, driving fidelity, temporal fidelity, and deployment behavior are different observables. A low image distance does not establish audiovisual synchronization, and high steady-state throughput does not establish low time to first frame (TTFF) or stable long-horizon operation. We therefore group evidence by what is measured and record the protocol that makes each value interpretable.

Four classes of evidence are useful. *Full-reference measures* compare an output with a paired and aligned target. *Distributional measures* compare sets of generated and real samples. *Task-specific evaluators* estimate an attribute such as face identity or audiovisual synchronization. *System and longitudinal measures* characterize the visible pipeline boundary and change over an evaluated session. These classes complement rather than replace one another.

8.3 Reference-Based and Distributional Visual Quality

Pixel errors, peak signal-to-noise ratio (PSNR), and the structural similarity index (SSIM) require a paired reference. SSIM compares local luminance, contrast, and structure before spatial aggregation [184], and its relationship to PSNR is itself protocol-dependent [67]. Its original validation concerns aligned still images; a video-level report must therefore state the frame pairing and temporal aggregation, and it should not interpret frame-wise SSIM as a temporal-consistency result. Learned perceptual image patch similarity (LPIPS) instead aggregates channel-normalized deep-feature differences across layers [214]; ensemble variants improve its robustness to transformations [84]. A lower value means that the candidate is closer to the paired reference under the selected backbone and calibration variant. Comparisons must retain the same network, layer set, preprocessing, crop, dynamic range, and aggregation.

The Fréchet Inception Distance (FID) fits Gaussian statistics to pretrained Inception features from real and generated image sets and computes their squared Fréchet distance [64]. Because this construction discards sample order, FID is an image-distribution measure rather than evidence of temporal stability. Fréchet Video Distance (FVD) applies the same statistical form to spatiotemporal features; the original study selected Kinetics-400 RGB Inflated 3D ConvNet (I3D) logits and showed that finite-sample values depend on sample count [170]. Consequently, an FID or FVD number is comparable only after the reference set, feature network and checkpoint, activation layer, preprocessing, sample count, and, for FVD, clip construction and content frame rate have been fixed. Neither distance isolates identity, lip synchronization, or deployment speed. The same protocol dependence applies to similarity scores computed with other pretrained encoders, such as CLIP-based text or image similarity [142]: the encoder checkpoint and preprocessing are part of the metric’s identity.

8.4 Human-Specific Fidelity

Human-specific evaluators require equally explicit boundaries. ArcFace is a face-recognition representation trained with an additive angular margin [43]; it is not itself a standardized virtual-human quality score. When a paper reports ArcFace-based cosine similarity (CSIM), the report should identify the embedding model and checkpoint, face detector and alignment, reference image or set, handling of failed detections, and frame or track aggregation. Recognition accuracy reported by the ArcFace source cannot be transferred into a threshold for generated-video identity fidelity.

For audiovisual synchronization, the underlying evaluator is SyncNet, which learned a joint audio–video embedding for lip-sync assessment in the wild [35]. Wav2Lip introduced lip-sync error distance (LSE-D; lower is better) and lip-sync error confidence (LSE-C; higher is better) on top of a pretrained SyncNet [138]; the evaluator differs from the synchronization expert used to train its generator. Cross-paper comparison therefore requires the same evaluator checkpoint, audio–video pairing, temporal window, face and mouth crop, preprocessing, and averaging rule. These scores test the selected audiovisual correspondence protocol; they do not measure expression quality, identity preservation, or latency.

Landmark Distance (LMD) was defined for paired lip-motion evaluation using Dlib detections on generated and ground-truth video [25]. The original construction uses 20 lip landmarks, aligns the mean landmark positions, computes corresponding Euclidean distances, and averages over points and frames; lower is better. LMD is thus a paired landmark-accuracy measure, not a no-reference audio–lip synchronization score. Detector version, landmark subset, crop and coordinate scale, alignment, frame correspondence, failure policy, and aggregation must remain fixed for numerical comparison.

Identity, synchronization, and landmark measures cover different failure modes. They should be paired with task-appropriate motion measures only when the underlying detector, normalization, reference, and aggregation are documented. A learned evaluator’s name alone is not a reproducible protocol.

8.5 Whole-Body, Interaction, and Cross-Scene Evaluation

Recent avatar systems expand the evaluation surface beyond face-only fidelity. Whole-body generation calls for separate face lip-sync, hand-keypoint consistency, gesture–audio alignment, body-motion diversity, face–body temporal coherence, and identity consistency

Table 15 Evidence scope and minimum protocol fields for representative visual-quality measures. PSNR denotes peak signal-to-noise ratio; SSIM, structural similarity index; LPIPS, learned perceptual image patch similarity; FID, Fréchet Inception Distance; and FVD, Fréchet Video Distance. The arrows $\uparrow$ and $\downarrow$ indicate that higher and lower values, respectively, are preferable only within a matched protocol.

Measure	Comparison unit	Reference	Minimum fields to report	Does not by itself establish
PSNR / SSIM $\uparrow$	Aligned image or frame pair	Paired target	Crop and alignment, color space, dynamic range, resolution, implementation, frame aggregation	Temporal coherence, identity, lip sync, or system timing
LPIPS $\downarrow$	Aligned image or patch pair	Paired target	Backbone, calibrated variant, layer set, normalization, crop, resolution, aggregation	Attribute-specific fidelity or temporal behavior
FID $\downarrow$	Image sets	Real image set	Reference set, feature checkpoint/layer, preprocessing, sample count, random sampling	Sample order, temporal stability, identity, or lip sync
FVD $\downarrow$	Video-clip sets	Real clip set	Video encoder/checkpoint/layer, clip length and sampling, frame rate, preprocessing, sample count	Failure localization, long-horizon stability, or runtime

across body regions. Listening-oriented systems additionally require response appropriateness, listener-motion diversity, speaking/listening transition smoothness, turn-taking responsiveness, and user preference; these should be reported separately from the latency of receiving the partner’s signal. Cross-scene identity evaluation should distinguish same-scene retention from changes in background, lighting, framing, or viewpoint, because a persistent reference may preserve identity in one scene while failing under a context shift.

Temporal wobble and tracking stability form another independent axis. A render-based system can preserve a fixed identity asset while introducing jitter through tracking, deformation, Gaussian mapping, or compositing. Reports should therefore specify the temporal artifact detector, reference coordinate system, time bins, failed-detection policy, and aggregation, and should not substitute a single-frame identity score or renderer FPS for a longitudinal wobble measurement. These dimensions are complementary to ArcFace, SyncNet, and LMD and should be stratified by anatomy, interaction mode, scene constancy, and failure type [1, 34, 92].

A complete protocol should disclose which of these measurements are paired, no-reference, or human-rated, together with the continuous duration and reset policy. At present, the literature reports subsets under incompatible settings rather than a unified whole-body, dyadic, cross-scene benchmark.

8.6 System Timing

System evaluation follows the boundary in Section 2.6. At minimum, a speed claim should state hardware and device count, output resolution and content frame rate, precision, batch size or concurrency, warm-up, measurement window, and every included component. Frames per second (FPS) is a frame-throughput measure only when frames are counted over a disclosed wall-clock generation boundary; otherwise an FPS value may denote

content or capture frame rate. Real-time factor (RTF) compares wall-clock time with generated-media duration; TTFF starts at a declared input-arrival event; and inter-chunk latency measures the time between successive committed chunks under a declared buffering policy. Number of function evaluations (NFE) records iterative model calls for a stated generated unit. FPS and RTF can be converted when frame count, media rate, elapsed interval, and boundary match (Section 2.6); neither determines TTFF or NFE. Measurements with different boundaries do not define a controlled cross-route ranking.

8.7 Longitudinal and Human Evaluation

Longitudinal evaluation measures how output quality and system behavior change during continuous generation. A reproducible protocol should specify session content and duration, initialization, retained context, and reset policy. Identity and appearance should be measured in predeclared time bins relative to an initial or short-horizon reference, alongside synchronization, motion, temporal artifacts, memory, and latency. Failures, resets, and recovery intervals require separate reporting because averages can conceal them. Ablating the active L1–L5 mechanisms under matched conditions would further distinguish their individual contributions from the behavior of the complete system.

We recommend human studies when automatic measures do not isolate visible artifacts or preference. Reports should state the question, stimuli, duration, and presentation order. They should also report rater recruitment, repeated judgments, aggregation, uncertainty, and the statistical test. A preference percentage without these fields is descriptive rather than a transferable quality scale.

Table 16 Minimum reporting record for Fast and Stable evidence. FPS denotes frames per second; RTF, real-time factor; TTFF, time to first frame; NFE, number of function evaluations; and NR, not reported in the primary paper.

Record	Required fields	Interpretation boundary
Measured frame throughput (FPS)	Frames counted, wall-clock elapsed interval, resolution, batch or concurrency, warm-up, hardware, precision, included pipeline	Establishes FPS throughput only for the disclosed generation boundary; does not establish latency or content/capture frame rate
RTF	Wall-clock numerator, generated-duration denominator, clip or stream length, pipeline boundary	Values below one establish faster-than-playback only for that convention
TTFF	Input-arrival event, buffering, first observable frame	First-frame service metric; does not establish steady-state inter-chunk latency or throughput
Inter-chunk latency	Successive committed-output timestamps, commitment interval, buffering, jitter summary	Steady-state service metric; does not establish TTFF or throughput
NFE / serial calls	Generated unit, solver steps, denoiser or vector-field calls, guidance multiplier, decoder calls	Call count does not establish wall-clock speed
Stability protocol	Evaluated continuous duration, persistent state, context bound, time bins, identity and appearance drift variables, failure counts, reset policy, and resource trajectories	A long demonstration or aggregate dataset duration is not quantitative Stable evidence

8.8 A Worked Portrait Evaluation Protocol

The following protocol is a proposed, implementable study design, not an experiment performed in this survey. Its primary task is single-reference, audio-driven portrait generation. Evaluate subject-specific methods in a separate enrollment stratum with a fixed capture budget; do not pool them with one-shot systems. Freeze checkpoints, code revisions, seeds, source images, audio files, and evaluator hashes in a manifest before running the models.

Inputs and continuous sessions. Select 20 held-out, consented identities and three seeds per identity. Use the same reference, audio, and seed list across eligible methods. Prepare two ten-minute audio conditions: a repeated 30-s passage as a controlled stress test, and continuous varied speech containing pauses, changes in speaking rate, and different phonetic content. Keep both audio manifests and their condition labels. For models with compatible pose or expression controls, add a separate change-following condition with scheduled turns and expression changes at 120, 300, and 480 s; record targets and transition intervals. Do not penalize audio-only models for unsupported controls or score an intended appearance change as identity drift.

Quality and change over time. Score non-overlapping 10-s bins over the full 600-s run, and report summaries for 0–60, 60–180, 180–300, and 300–600 s. Retain both absolute quality and change from the first valid bin. Identity uses a fixed reference embedding and declared face alignment; synchronization uses a frozen SyncNet implementation and its declared audio/video windows. For motion-controlled paired trials, report pose or landmark target error and boundary jumps. For audio-only motion, report motion magnitude and diversity separately from lip synchronization; low motion alone is not stability. Add a perceptual quality assessment or

blinded artifact rating so that a consistently poor or frozen output cannot pass solely by exhibiting little drift.

Failures and recovery. Log every missing face detection, invalid landmark track, dropped output, visible corruption, stall, and reset. Invalid detections remain in the denominator as a reported failure fraction; compute conditional metric summaries on valid frames and disclose their coverage. The primary run allows no manual reset. A second, explicitly labeled recovery run may use a pre-declared policy; report reset count, downtime, and time to return to the accepted quality range. Publish the range and its calibration procedure before examining method results. Runs that terminate early contribute their observed duration and failure event, rather than disappearing from the average.

Timing and resource traces. Stream audio at its playback rate. Start the TTFF clock when the first input packet arrives; stop it at the first observable RGB frame. Separately log input accumulation, encoders, visual generation, decoder/rendering, composition, transfer, and presentation timestamps. Report cold and warmed starts, total completed frames per wall-clock interval, inter-commit median and 95th percentile, stalls, GPU allocation, and peak/resident memory once per second. Synchronize device timers when measuring components and retain a wall-clock service trace to account for overlap. Report enrollment time and resources separately. Test a fixed single-stream configuration first; concurrency is a separate experiment.

Uncertainty and interpretation. Report per-identity results and paired method differences. Bootstrap identities, retaining all their seeds and time bins together, for 95% confidence intervals; frames within one run are not independent samples. Stratify repeated versus varied input

Table 17 Task–dataset–measurement map. Dataset availability does not establish a ready-made long-session split; the session manifest and reuse permissions must be specified. Whole-body entries indicate selected extensions to the portrait-centered scope.

Task	Candidate source material	Measurements and limits
Audio-driven portrait	HDTF; MEAD	Identity, lip synchronization, expression/control fidelity, artifact ratings, and continuous service traces. Paired image metrics require aligned targets.
Video-driven reenactment	VFHQ; CelebV-HQ	Pose/landmark target error, identity, temporal artifacts, and disocclusion. Unpaired cross-identity reenactment lacks a pixel-matched RGB target.
Subject-specific rendering	NeRSemble; declared subject capture	Novel-view/reenactment quality, tracking/deformation wobble, capture and fitting cost. Multi-view capture is an enrollment condition.
Selected upper/full-body animation	OpenHumanVid; task-specific video/pose sequences	Face synchronization, body/hand keypoint fidelity where annotations permit, gesture–audio correspondence, and face–body consistency. Face identity scores do not cover hands or body motion.

and fixed versus changing controls. A mechanism ablation uses identical inputs, hardware, and serving settings unless the changed resource is itself the object of the experiment. Ten minutes is the target of this proposed study, not a prerequisite for including historical methods in the survey. Shorter supported horizons remain valid when clearly labeled.

9 Discussion, Challenges, and Future Directions

The four routes offer different ways to distribute appearance, motion, and temporal state across a generation pipeline. Their practical value depends on the task, available resources, and evaluation conditions. The evidence reviewed in Sections 3–8 motivates four common research problems: composing modules with explicit interfaces, measuring complete-system cost, maintaining consistency with bounded state, and evaluating deployment safeguards.

The synthesis in Section 7 separates measured effects from architectural deductions. The following questions build on those findings.

9.1 Composition Requires Explicit Interfaces

Hybrid systems often divide work among a condition or motion model, a visual representation owner, a decoder or renderer, and a compositor. Under the ownership rule in Section 2.3, the route label follows the module that produces or persistently stores dense visible content [21, 60]. Secondary modules remain part of the system boundary. Replacing one module can change motion control without changing the visual owner; it can also reduce visual-model calls while leaving condition encoding, decoding, and transport on the critical path.

Composition requires both a representation interface and a state-update policy. Each module should specify its coordinate system, identity reference, subject-specific

assets, retained history, reset behavior, and recovery procedure. These fields determine when output becomes available and which errors can propagate. For example, Teller’s audio buffering precedes motion prediction and rendering, while LivePortrait’s reused appearance still requires motion estimation and decoding [58, 226].

Compact motion interfaces can reduce the state presented to a dense visual owner, but they also restrict appearance changes to those expressible by the interface. Dense latent interfaces offer more freedom while making identity ownership and failure localization harder to audit. These are testable architectural trade-offs rather than universal advantages of a route. A falsifiable research question is whether a hybrid can expose typed representation, state, and recovery contracts such that the latency and failure contribution of every module is reproducible in isolation and in composition.

9.2 Complete-System Cost and Timing

The source literature reports FPS, wall-clock latency, real-time factor, function evaluations, and relative speedups under different boundaries. Tables 3, 5, 8, and 10 preserve those route-specific boundaries. A renderer-only measurement is not an end-to-end system result [58, 76], and source values remain interpretable only with their original hardware, precision, concurrency, warm-up, generated unit, and included components, as formalized in Section 8.6.

A deployment record should account separately for subject enrollment or asset preparation. The runtime ledger should time condition acquisition and encoding, motion or state generation, dense visual synthesis, decoding or rendering, compositing, transfer, and I/O. User-visible measures include time to first frame, inter-commit latency and jitter, steady throughput, memory, and recovery cost. The critical path moves with the operating point. The within-source analyses in Section 7 illustrate two consequences: decoder choice can change

both throughput and output quality, and multi-device pipelining can raise throughput while increasing the wait for the first committed output. Component timings are needed to establish which stage is the bottleneck. A falsifiable research question is which component owns the critical path at each resolution, chunk size, concurrency level, and session duration, and whether the same conclusion survives when enrollment, I/O, and recovery are included.

9.3 Bounded-State Long-Horizon Generation

The five patterns in Section 2.7 provide a common vocabulary for mechanisms whose evaluation remains uneven. The reviewed diffusion systems contain examples of all five, including Live Avatar’s sampled observations beyond an hour under looped audio [76]. Render-based systems reuse an identity asset but can drift through their controllers or rendering pipeline; DyStream’s 180-s pose-drift ablation provides a limited longitudinal example [21]. Native visual AR explicitly retains history, yet the reviewed systems lack a complete joint stability protocol. Framewise GAN generators avoid recursive visual feedback when earlier outputs are not reused, but can still exhibit jitter or inherit controller errors.

This comparison suggests studying which mechanisms transfer across representations. Training with generated histories could be evaluated in both native visual AR and recurrent motion controllers. Reliability-aware memory updates could be compared with fixed windows, compression, and periodic resets. A matched study should isolate each mechanism’s effect on identity, synchronization, motion diversity, and legitimate scene changes over predeclared time bins. The current evidence neither establishes a universal ordering of the routes nor supports unbounded-duration stability beyond the conditions actually evaluated.

Audio-to-deformation and expanded Stable axes. Recent audio-driven Gaussian systems expose a shift from the conventional audio $\rightarrow$ pose or SMPL $\rightarrow$ renderer cascade toward direct audio $\rightarrow$ Gaussian or particle deformation $\rightarrow$ renderer paths. This can preserve high-frequency facial, hand, and body dynamics that an intermediate pose interface may quantize, but it moves complexity into learned deformation fields and their tracking boundary. The resulting Stable problem is multidimensional: appearance identity, motion and deformation, audio–visual synchronization, interaction response, and system/resource behavior can fail independently. Generalizable Gaussian priors reduce enrollment cost but introduce adaptation and out-of-distribution trade-offs;

fixed Gaussian scenes anchor appearance but do not prevent wobble, tracking failure, or compositing drift. These systems make the missing evidence concrete: future reports should pair enrollment and runtime ledgers with time-binned identity, motion, synchronization, wobble, failure, and recovery measurements [1, 3, 69, 92, 104].

9.4 Route Selection Under Deployment Constraints

Table 18 summarizes overlapping suitability conditions. The design guidance of Sections 3–6 supports a first set of conditional guidelines. These compare route-level mechanism maturity under stated constraints; they do not rank individual methods, and each guideline inherits the measurement caveats of its underlying sources.

When per-subject enrollment is acceptable, render-based systems offer reusable appearance and reported high-throughput or wearable configurations [24, 188]. Selection should account for capture and fitting cost, followed by tracking, deformation, and compositing over the intended session duration.

When one-shot operation is required, compact motion generation over reused appearance provides a practical design pattern [58, 63, 192]. The motion module and renderer must both fit the latency budget, while deformation and disocclusion determine which movements the appearance model can reproduce.

When changes extend beyond a reusable appearance model, dense diffusion offers mechanisms for synthesizing broader visual content. The reviewed streaming examples combine distillation, memory, and substantial serving resources [76, 152]. Persistent anchors can also restrict scene changes, so the intended variability must be evaluated directly. Component profiling should determine whether further optimization targets the sampler, decoder, or conditioning path.

When the edit is confined to a small region, feed-forward GAN synthesis can reduce the amount of generated content [138, 218]. Evaluation should include compositing seams, synchronization, and framewise jitter as well as runtime.

When explicit visual-history modeling is central, native visual AR offers a direct formulation. Its deployment evidence is less complete in the reviewed sources: Table 8 does not contain a system with both full-pipeline timing and a joint time-binned stability evaluation. The representation, local emission head, and state policy should therefore be validated for the intended application.

Table 18 Overlapping suitability conditions. D denotes documented capability; H denotes a proposed deployment hypothesis requiring task-matched validation.

Requirement	Candidate mechanism / status	Additional condition to check
Regional lip editing	Wav2Lip+GAN, MuseTalk; D	Source pixels remain usable; inspect seams, synchronization, and jitter.
Reusable identity and controllable pose	Fields, Gaussians, feature warps; D	Enrollment is feasible <i>and</i> the representation supports the intended appearance, views, and motion.
One-shot portrait animation	LivePortrait, VASA-1, LAM; D	Test disocclusion, pose limits, and complete controller-renderer timing.
Appearance or scene changes	Dense visual generation; H for a new deployment	Validate the specific change. A static-scene system such as Live Avatar does not establish open-ended scene editing.
Explicit visual-history control	Native visual AR; D for architecture	Profile local sampling and decoding; test long-rollout errors and resets.

9.5 Trustworthy Deployment Requires Layered Evidence

Fast and Stable systems sharpen deployment risks: low-latency synthesis enables live impersonation, and long-horizon stability with cached identity state extends it. Trustworthiness is not an additional scalar metric. The NIST AI 100-4 report separates provenance-data tracking from synthetic-content detection and discusses watermarks, metadata, filtering, and red-teaming as controls with different lifecycle positions and failure surfaces [19]; it provides no avatar-specific effectiveness or runtime measurement. The C2PA specification defines signed claims, content bindings, and validation states, including segment-level paths for live video [38]. These mechanisms bind a signing credential to covered media bytes and expose specified integrity failures; they do not establish factual truth, verify the depicted identity, or prove authorization to use a likeness, and they carry manifest-stripping, key-management, and privacy risks [36, 37].

Empirical safeguards have equally important boundaries. DeepfakeBench exposes generalization gaps across manipulation types with frame-level evidence [195], and Video Seal demonstrates faster-than-playback video watermarking under a stated offline boundary without adaptive-attack coverage [52]. A watermark hit, a detector score, and a valid provenance record are different observations; none alone proves authorization or harmlessness. A deployment evaluation should declare the attacker model, benign platform transformations, operating point, authorization and revocation procedures, and the latency cost of each control on the serving path. The last point connects trust directly to this survey’s Fast accounting, since provenance signing and watermarking must fit inside the same per-chunk budget the route mechanisms optimize.

9.6 Conclusion

Fast and Stable virtual-human generation depends on where appearance, motion, and temporal state reside and how the complete pipeline is served. The hybrid audit makes these choices explicit without treating the four editorial routes as disjoint learning objectives. Within-source evidence shows that a faster decoder can alter identity and temporal quality, additional serving parallelism can affect throughput and startup differently, and a beneficial short-horizon ablation does not establish long-session reliability. Reusable appearance narrows synthesis work while leaving motion, geometry, and composition to be tested. The source-linked supplement and proposed evaluation protocol turn these findings into reproducible questions about operating conditions, failure types, and the duration over which a claim is supported.

Data Availability Statement

This survey does not introduce a new video dataset. The accompanying evidence supplement, snapshot VH-2026-10-09, records the checked source editions, extraction locations, configuration boundaries, and unresolved fields. The literature catalogue is a selected, maintained reading resource rather than a complete index of the bibliography. Its frozen snapshot contains 164 records, including foundational and contextual references. The [public research companion](#) provides the frozen evidence downloads and catalogue snapshot VH-2026-10-09.

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